

The FTD Ontological Chain

A Canonical Map and Intuition Driver

2026

How to Read This Document

This document maps the complete six-layer ontological chain of Foundational Ternary Dynamics (FTD)—

ONTOLOGY > LOGIC > MATHEMATICS > PHILOSOPHY > PHYSICS > SCIENCE

—in the order the constitution declares it (FTD-0254). It is a SYNTHESIS document: it re-states existing canonical claims at their canonical tags and *introduces no new mathematics*. Its purpose is to be read *map-first*: an external reader should be able to take in the whole chain from the Master Chain DAG (Figure 1) and then walk each layer to see the one move it makes and exactly how far that move is established.

The reading discipline. Each layer is *constructed from* the layer before it and *accountable to* the layer after it: mathematics is built on the ontology and logic that precede it; physics must answer to the science (measurement) that follows it. Every load-bearing claim below carries an epistemic tag and, where one exists, its provenance identifier (FTD-NNNN), so the claim can be traced back into the project's ledgers. The tags are where the honesty lives; read them first.

THE PRECEDENCE RULE

Where this document disagrees with the LEDGER (LEDGER.md), the constitution (SPEC_FTD_FRAMEWORK_V1.md), or the algebraic spine (SPEC_ALGEBRAIC_SPINE.md) on any tag or value, **those documents are correct and this one has a bug**—please file it. The ordering is LEDGER > constitution > this document.

How epistemic status is shown. Status is carried *without colour*, by rule style alone, so it survives grayscale and monochrome print. A heavy rule is FORCED; a thin rule is CONDITIONAL; a dashed rule is CONJECTURAL; a double rule is a BOUNDARY—a mapped wall or an open kernel. The same vocabulary governs the DAG: a double-ringed node is an axiom or a framework declaration, a solid box is theorem-grade, a dashed box is below theorem grade, and a double-ruled box is a boundary.

THEOREM

FORCED — rigorously established (heavy rule)

PARAMETRIC

CONDITIONAL — calibration / selection (thin rule)

CONJECTURE

CONJECTURAL — evidence, no derivation (dashed rule)

OPEN

BOUNDARY — a mapped wall / open kernel (double rule)

The three registers. FTD separates three kinds of foundational input. Keeping them apart is what lets the framework be honest about what is derived and what is chosen:

Register	Tag class	Function
Postulates P1–P5	AXIOM	Generative. Discrete space, discrete time, ternary states (J -primary), Moore locality, determinism. Frozen; machine-checked.
Framework commitments FC-0, FC-1, FC-2	AXIOM (class)	Model-fixing declarations. Where a theorem proves a fork is open, a commitment picks the branch—the theorem proves the fork, the commitment picks the side. Not derivations. (FTD-0254, FTD-0255, FTD-0256.)
Calibrations ($a_{\text{phys}} \equiv \ell_P$, $M_{\text{REST}} = m_e$, t_{phys})	declared	Unit-fixing. Theorem-enforced as non-derivable (FTD-0059, FTD-0096). Dimensionless predictions are calibration-invariant; dimensional ones are calibration-conditional.

With the registers in hand and the DAG in view, the chain reads top to bottom.

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1. Ontology — What Exists

The move this layer makes

Every framework has to answer one question before it can say anything else: *what is there?* The ontology layer is FTD's answer, and the answer is deliberately spare. There is a three-dimensional cubic lattice with no completed boundary. On that lattice live two fields, and only two: a continuous vector *flux* that says what each site is *disposed* to do, and a discrete ternary *state* that records what each site has *actually* done. Time is a separate primitive — a global tick, not a fourth axis of space. Nothing else is fundamental: charge, mass, particles, and even the metric of spacetime are things the framework hopes to *read out* of this substrate later, not things it puts in by hand.

The intuition to carry forward is the split between *disposition* and *actuality*. The flux J is a field of tendencies; it is smooth, it carries waves, and in isolation it runs forward and backward equally well. The state s is the ledger of commitments the lattice has made; it takes only three values, and once a commitment is made the bookkeeping that made it cannot in general be undone. These two fields do not blend. They touch through exactly two doors — one that lets a built-up disposition *become* a state (genesis), and one that lets a committed state *source* the disposition of its neighbours (Gauss). Holding those two doors in mind is most of what it takes to follow the rest of the chain: the mathematics of Chapter 3 is the arithmetic of the lattice's symmetry, and the physics of Chapter 5 is the IR readout of these same two fields run forward in time.

A reader should also note what this chapter does *not* claim. It does not claim that the two-field split is forced; that is a modelling choice, and it is tagged as one. It does not claim that spacetime is built; the metric is promised to Chapter 5, and at this layer space and time are simply two separate primitives. The honesty of the chain lives in those tags, so they are stated inline as each claim appears.

1.1. The five postulates

The substrate of FTD is fixed by five postulates, frozen and machine-checked, each an **AXIOM** (the constitution, FTD-0254, §1.1):

P1 — Discrete space.

A three-dimensional cubic lattice of voxels with *undefined boundary*: at every specified position the axis-adjacent sites exist, but the lattice is not a completed-infinity $\mathbb{Z}^3$ totality. Arbitrarily large finite computations are well posed; completed limits are not. **AXIOM**

P2 — Discrete time.

Dynamics proceeds in global ticks. Simultaneity at the substrate level is absolute — a

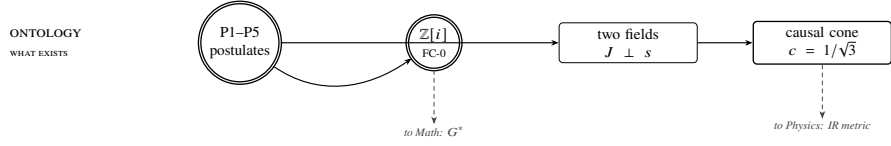


Figure 1.1.: This chapter in the chain. The four load-bearing nodes of the Ontology layer: the five postulates and the **AXIOM**-class $\mathbb{Z}[i]$ reading (FC-0) are seeds (double-ringed); the two-field decomposition $J \perp s$ is a selection/synthesis (thin box); the causal cone $c = 1/\sqrt{3}$ is theorem-grade (solid box). Solid arrows read “constructed from”; the greyed arrows point at what later layers build on this one.

privileged foliation, the neo-Lorentzian reading made explicit by FC-2 in Chapter 2. **AXIOM**

P3 — Ternary states, J -primary.

Each voxel carries a continuous flux vector $J \in \mathbb{R}^3$ (dispositional) and a ternary state $s \in \{-1, 0, +1\}$ (actual). The flux J is *primary*; the state s is the manifestation projection of J through the genesis threshold rule (§1.3). Treating s as primary would double-count. **AXIOM**

P4 — Local causality.

Updates depend only on the 26-neighbour Moore neighbourhood at the previous tick; information moves at most one voxel per tick. **AXIOM**

P5 — Determinism.

The update map is a function. *Determinism is not reversibility*: a deterministic map may be many-to-one (the point Chapter 2 turns into the arrow of time). **AXIOM**

These five are the only generative inputs. Everything the framework derives is quantified over exactly P1–P5; the machine-checked theorems of the corpus (FTD-0243, FTD-0253) take precisely this set as their hypothesis.

1.2. Graded monism and the $\mathbb{Z}[i]$ reading (FC-0)

The ontology is *graded monism*: one substance — the void — whose *dispositions* are the flux J , whose *manifestations* are the actualized states ± 1 , and whose *properties* (charge, mass) are emergent from manifestation patterns. The state field is not a second substance; it is the actualized manifestation of the flux under the genesis rule. This is the math-first ordering of FTD-0153: *ontology = finite mathematical invariant structure; physics = operationally stable readout of that structure*.

One modelling choice is promoted to a framework commitment because the entire algebraic spine of Chapter 3 rests on it:

FC-0 — THE $\mathbb{Z}[i]$ READING **AXIOM** (CLASS)

The cubic lattice’s order-4 planar symmetry is read as the arithmetic of the Gaussian integers $\mathbb{Z}[i]$, with unit group $\mathbb{Z}[i]^\times$ of order 4. *This is a declaration, not a derivation*. The

algebraic spine is forced by P1–P5 *together with* FC-0, not by the discrete ontology alone — the honesty correction installed by the construction monograph’s red-team (FTD-0249), here canonized as the zeroth framework commitment (FTD-0254). Declaring it does not derive it.

AXIOM

FC-0 belongs to the second of FTD’s three registers — the *framework commitments*, which sit between the generative postulates and the unit-fixing calibrations. The pattern repeats for FC-1 and FC-2 in Chapter 2: where a theorem proves the postulates leave a fork open, a commitment picks the branch FTD asserts the world is on. The theorem proves the fork; the commitment picks the side.

1.3. Two orthogonal fields, two coupling channels

FTD’s field content is *two orthogonal fields, nested* (the constitution §1.3, FTD-0257). The decomposition is canonical but it is a *choice* of how to package the dynamics, not a new dynamical claim — hence its tag is a synthesis-and-selection, never a theorem:

Claim 1.1 (The primary pair, $J \perp s$). The primary field content is the orthogonal pair

$$J \in \mathbb{R}^3 \perp s \in \{-1, 0, +1\}.$$

J is continuous, reversible in isolation, and carries the wave dynamics; s is discrete, finite, and irreversible (manifestation is many-to-one). They are *orthogonal* in the precise sense that they inhabit different ontological grades — disposition versus actuality — and interact through exactly two channels and no others. [SYNTHESIS] + [SELECTION] (FTD-0257)

Why “*orthogonal*.” There is no state–state dynamics and no flux–flux manifestation; the two fields meet only at the two doors below. The word is structural, not metric: it names the absence of any third coupling channel.

Channel 1 — Genesis / evaporation (downward, $J \rightarrow s$).

At a void site, $|J|$ crossing the manifestation threshold projects a state (the s = sign rule); falling below the evaporation threshold reverts it. The kinetics scale is K_{MANIFEST} and the trigger is $K_{\text{GENESIS}} = N_c \cdot K_{\text{MANIFEST}}$. This is the engine’s `phase_write` step. It is the *many-to-one* door: it is where irreversibility, and therefore the arrow of time of Chapter 2, enters.

Channel 2 — Gauss sourcing (upward, $s \rightarrow J$).

Manifested states source the flux through the lattice Gauss constraint

$$\nabla \cdot J = \rho_s,$$

where ρ_s is the manifested charge density. This is the engine’s `gauss_project` step. It is the *linear* door: it feeds committed states back into the dispositions of their neighbours, and it is the origin of the geometric Coulomb law of Chapter 5.

Everything else in the framework is dynamics *within* one field or the other. The two channels are the engine’s actual architecture; the constitution packages them as the canonical ontology, and that packaging is the SELECTION above.

1.3.1. The nested pair: symplectic quadratures (q, p)

Inside the flux layer there is a second, nested orthogonality. The flux sector is a second-order field, so its true arena is phase space: per mode, the amplitude q and its rate $p = \partial_t J$ form a symplectic pair (q, p) . The substrate’s only native dynamical angle is the quadrature phase $\arg(q + ip)$, winding at the dispersion frequency derived in §1.5. This pair carries a distinction that becomes load-bearing in Chapter 2:

Claim 1.2 (The Crucial Distinction). The quadrature pair is symplectically non-trivial yet observably commutative:

$$\{q, p\} \neq 0 \quad \text{yet} \quad [q, p] = 0$$

— a non-zero Poisson bracket but a vanishing observable commutator (FTD-0243). The quadrature *clock* is the substrate’s native time-keeper; identifying it with “the measurement angle” is interpretive SELECTION, and the incompatibility that would make choosing a quadrature a quantum measurement is exactly what the framework declines to import (FC-1, Chapter 2).

The two orthogonalities live at different levels and do not compete: the primary pair is orthogonal at the *grade* level (two coupling channels only); the nested pair is orthogonal in the *symplectic* sense (conjugate quadratures of one oscillation, internal to the flux member of the primary pair). Chapter 2 reproduces the argument behind Claim 1.2 in full; here it is recorded as part of the field inventory.

1.4. Space $\perp$ time at the base

P1 (the lattice) and P2 (the tick) are *separate primitives*. FTD’s base ontology has a 3-space and a 1-time, not a 4-geometry. Nothing at the postulate level mixes them (the constitution §1.5):

Claim 1.3 (Separation at the base). At the postulate level, space and time are orthogonal primitives. The *causal cone* that relates them is a theorem of locality (§1.5); the Lorentzian *metric* that would weave them into Minkowski spacetime is *not* forced by P1–P5 (FTD-0253). FTD’s committed reading — the mixing is emergent and sector-scoped, the separation fundamental — is FC-2, declared in Chapter 2 and measured in Chapter 5. AXIOM (base structure) + FC-2 forward-pointer

This is the single most important orientation for an outsider: at the ground floor of FTD there is *no spacetime*. There is space, there is time, and there is a maximum speed at which one can influence the other. That maximum speed is the one thing about the space–time relation that the postulates *do* force, and it is derived next.

1.5. Reproduced derivation I: the causal cone $c = 1/\sqrt{3}$ and the lattice dispersion

The single number this layer forces about propagation is the causal speed. It follows from the CFL (Courant–Friedrichs–Lewy) stability condition for a discrete wave equation on the cubic lattice, with no appeal to the state space (the constitution §3.2 / §5.1 row 1; FTD-0253; FOUND_AXIOM_ZERO.md §2.2(d)).

Result 1.1 (CFL causal cone). On a D -dimensional cubic lattice with unit voxel spacing and unit tick, the maximum stable propagation speed of the flux wave is

$$c = \frac{1}{\sqrt{D}}, \quad \text{so at } D = 3 : \quad c = \frac{1}{\sqrt{3}}.$$

THEOREM (FTD-0253)

Why $1/\sqrt{D}$. Take the second-order flux wave equation that governs the dispositional field, discretized on the lattice by the standard leapfrog/centred scheme. With voxel spacing a and tick Δt set to unity, the update for a mode of wavevector $k = (k_1, \dots, k_D)$ replaces each second derivative by a centred difference. The temporal difference contributes a factor $-4 \sin^2(\omega \Delta t/2)$ and each of the D spatial differences contributes $-4 \sin^2(k_j/2)$. Matching the two sides of the discretized wave equation gives the dispersion relation

$$\sin^2\left(\frac{\omega}{2}\right) = c^2 \sum_{j=1}^D \sin^2\left(\frac{k_j}{2}\right).$$

For the scheme to be stable — for ω to stay real for *every* mode, including the worst case where all $\sin^2(k_j/2) = 1$ simultaneously (the lattice corner $k_j = \pi$) — the right-hand side must never exceed 1. The worst case sums to D , so stability requires

$$c^2 \cdot D \leq 1 \quad \implies \quad c \leq \frac{1}{\sqrt{D}},$$

and the maximum stable speed saturates the bound. At $D = 3$ this is $c = \frac{1}{\sqrt{3}}$. This is a theorem about discrete wave equations on the lattice; it does not depend on the ternary state space at all — it is purely the geometry of P1 plus the one-voxel-per-tick locality of P4.

The lattice dispersion. In the long-wavelength (small- k) regime where a single spatial mode dominates, the same matching condition reduces to the FTD flux-wave dispersion

$$\omega(k) = 2c |\sin(k/2)|,$$

with $c = 1/\sqrt{3}$. The frequency $\omega(k)$ is the rate at which the quadrature phase $\arg(q + ip)$ of §1.3.1 winds; it is the substrate's native clock. Two features matter downstream. First, for small k the relation linearizes to $\omega \approx c |k|$, recovering an ordinary light-cone in the IR — the seed of the

emergent metric promised to Chapter 5. Second, at large k the sin bends the dispersion away from linearity: this is a *native* ultraviolet departure from exact Lorentz invariance, not a defect, and it reappears as a falsifiable prediction in Chapter 6.

Note what has and has not been built. The cone — the fact that there is a finite maximum influence speed — is **THEOREM** and holds for *any* local lattice dynamics. The Lorentzian *metric* that would promote the cone to a full spacetime geometry is not in hand here; per Claim 1.3 it is emergent and sector-scoped, and the work of building it is Chapter 5's.

1.6. Reproduced derivation II: the genesis / Gauss coupling structure

The two coupling channels of §1.3 are the only places the two fields meet, and their structure is fixed by the postulates. We state each as a rule on the configuration (s, J) , making explicit why one is irreversible and one is linear.

The genesis projection (downward, $J \rightarrow s$). Genesis is a threshold map applied at each void site v ($s(v) = 0$) once per tick, after the flux update:

$$s(v) \mapsto \begin{cases} \text{sign}(J(v) \cdot \hat{n}), & |J(v)| > K_{\text{GENESIS}}, \\ 0, & |J(v)| < K_{\text{MANIFEST}} \text{ (evaporation)}, \\ s(v), & \text{otherwise (hysteresis band)}. \end{cases}$$

This is the engine's `phase_write` step, with trigger $K_{\text{GENESIS}} = N_c \cdot K_{\text{MANIFEST}}$. Three properties are load-bearing. (i) It is a *function* of the local flux, so it preserves P5 (determinism). (ii) It is *many-to-one*: distinct flux histories that cross the same threshold with the same sign land on the same state, and the hysteresis band makes the inverse genuinely undefined — so the map is provably *non-invertible*. This many-to-oneness is the finite, irreversible content of the substrate, and it is what Chapter 2 elevates into the native arrow of time (FC-2). (iii) The threshold K_{MANIFEST} is the manifestation quantum; it is fixed in the *calibration* register ($M_{\text{REST}} = m_e$), not derived here.

The Gauss sourcing (upward, $s \rightarrow J$). The committed states source the flux through the lattice Gauss constraint, imposed once per tick by projection:

$$\nabla \cdot J = \rho_s, \quad \rho_s(v) = (\text{manifested charge at } v).$$

Concretely, the projection step replaces J by the nearest field whose lattice divergence equals the manifested density ρ_s — a linear (Poisson) solve against the manifestation pattern. Two properties matter. (i) It is *linear* in the source, so superposition holds and the resulting static potential is the lattice Poisson Green's function — exactly the geometric Coulomb law reproduced in Chapter 5. (ii) It runs *upward* only: states set boundary data for the flux, but the Gauss step itself never creates or destroys a state. The asymmetry between the two channels — one many-to-one and

irreversible, one linear and reversible — is the ontological seed of every later asymmetry in the chain, including the π/G^* split of Chapter 3 and the arrow of Chapter 2.

Together, Results 1.1 and the two channel rules close the ontology: a finite ternary lattice, two orthogonal fields, one maximum speed, and two doors between disposition and actuality. Nothing else is fundamental.

1.7. What this layer does not settle

The ontology is deliberately spare, and its honesty is in naming what it leaves unbuilt:

- **The metric is not built here.** The causal cone is **THEOREM** (Result 1.1), but the Lorentzian metric that would make space and time into one geometry is *not* forced by P1–P5 (FTD-0253). Whether the substrate carries an emergent IR metric is committed by FC-2 (Chapter 2) and measured, sector-scoped, in Chapter 5 — it is not a result of this layer.
- **The two-field decomposition is a choice, not a theorem.** The primary pair $J \perp s$ and its two-channel coupling are packaged as canonical ontology, but that packaging is **SELECTION** + **SYNTHESIS** (FTD-0257), not a forced decomposition. A reader is entitled to ask whether a different bookkeeping of the same dynamics is possible; the framework’s answer is that this one is canonical, not that it is unique.
- **FC-0 is declared, not derived.** The reading of the lattice symmetry as $\mathbb{Z}[i]$ (§1.2) is an **AXIOM**-class commitment. The whole spine of Chapter 3 is conditional on it, and the condition is stated, not proven. What would overturn FC-0 is set out in Chapter 6’s kill-conditions.
- **The native angle is a selection.** Identifying the quadrature clock of §1.3.1 with a “measurement angle” is interpretive **SELECTION**; the framework records the clock as the native time-keeper and declines, at the next layer, the import that would make reading it a quantum measurement.

With the substrate fixed — a finite ternary lattice, two orthogonal fields, the causal cone, and the two coupling channels — the chain proceeds to ask what *follows* from these commitments, and what the framework deliberately declines to import. That is the work of Chapter 2.

2. Logic — What Follows, What Is Declined

The move this layer makes: a classical world that declines the quantum import

Ontology said *what exists*: a finite ternary lattice carrying two orthogonal fields, with a forced causal cone. Logic asks the next question: given those things, *what follows*, and—just as importantly—*what does the framework decline to assume*? This is the layer where FTD takes its sharpest, most consequential positions, and it takes them in the open.

The move is this. The substrate, run forward under its five postulates, exposes a world whose observables behave *classically*: every quantity you can read off a configuration is an ordinary real-valued function, and ordinary functions multiply commutatively. There is no hidden non-commutativity waiting to be distilled by a clever limit. Quantum mechanics' defining feature—observables that fail to commute, so that measuring one disturbs another—is therefore not something the substrate *contradicts*; it is something the substrate simply does not contain. FTD's response is not to manufacture it and not to deny it, but to *decline* it: to commit, as a matter of framework definition, that the commutative substrate is the whole of its ontology of observables.

That single move has a delicate companion. The substrate's dynamics are far from trivial—the leapfrog time-step gives its phase space a genuine symplectic structure, the same structure that in classical mechanics generates Hamiltonian flow. So one is tempted to say “but surely a non-trivial symplectic structure *is* quantum-like.” The whole of this chapter's reproduced derivation exists to defuse that temptation precisely. There are *two* brackets at play, and they are different objects: the Poisson bracket lives on the *functions* and is non-zero; the observable commutator lives on the *algebra's product* and is zero. The substrate is symplectically rich and algebraically commutative at the same time, with no contradiction. The quantum deformation that would turn the first bracket into the second requires an external parameter the substrate does not supply.

A second decision rides on the same logical base. P5 makes the update a *function*, not an invertible map—determinism is not reversibility—and the genuinely finite part of the substrate, the ternary state field, is provably many-to-one. So the framework declines global time-reversal symmetry as well, taking the arrow of time as native rather than emergent. Two declinations, one classical-logical foundation. Everything below states each at exactly the epistemic grade the ledgers record: the algebra result is a theorem; the two declinations are *declarations*, not derivations; and one piece of the observer story is honestly left open.



Figure 2.1.: This chapter in the chain. The Logic band: the substrate exposes a strictly *commutative* observable algebra A_5 (theorem-grade), on the strength of which two framework commitments are *declared*—FC-1 declines the quantum measurement import, FC-2 takes the arrow of time as native—and one kernel stays open (the *sharpness* of the observer’s blind spot). Shapes and rules carry epistemic status exactly as in Figure 1.

2.1. Determinism is not reversibility

Postulate P5 states that the update map is a deterministic function of the configuration. It does *not* state that the map is invertible. A deterministic map may be many-to-one—heat diffusion is the canonical example of a process that is deterministic *and* irreversible. This distinction is not a technicality; it carries the entire weight of FC-2 below (§2.5). SYNTHESIS (FTD-0256, licensing boundary FTD-0253).

The point to fix is that the postulate base *permits* an arrow without yet *committing* to one. Whether reversibility is a property of the world is, at the level of P1–P5, an open fork; §2.5 records which branch FTD takes and why that is a declaration rather than a theorem.

2.2. The substrate’s observable algebra is commutative

Let the *observable algebra* A_5 be the smallest collection of real-valued functionals of the configuration $(s, J, \text{wave_vel}, \mathcal{L})$ that contains those generators and is closed under pointwise real-linear combination, pointwise product, Moore-neighbourhood sums, and composition with the deterministic update map U . The central logical fact of the substrate is:

Result 2.1 (Commutativity of A_5). The observable algebra A_5 is strictly commutative.

THEOREM (FTD-0243; algebraic core machine-checked in `lean/StandAlone.lean`). A commutative algebra has a distributive (Boolean) event lattice by the Birkhoff–von Neumann correspondence, so a joint probability distribution over its observables *always* exists. In a word: **the substrate’s logic is classical logic**. Non-contextual hidden variables are not an extra hypothesis here—they are forced by the algebra. The full argument, and the crucial Poisson-versus-commutator distinction it turns on, is reproduced in §2.7.

2.3. The four closed routes to non-commutativity

Result 2.1 is an *absence* statement, and absence statements earn their keep only by surviving attempts to circumvent them. Each known FTD-native route to manufacturing quantum non-commutativity from P1–P5 has been tested and closed:

Route	Why it closes	Status
Modular time — a Tomita–Takesaki type III ₁ algebra read off the substrate	The substrate algebra is commutative $\rightarrow$ type I $\rightarrow$ trivial modular flow	CLOSED NEGATIVE FTD-0225
Manifestation as measurement — genesis + Gauss back-reaction read as collapse	A deterministic function of commuting flux gives a Boolean event lattice, i.e. classical coarse-graining	CLOSED NEGATIVE FTD-0226
Budget symmetry — the body-diagonal C_3 (the $N_c = 3$ count) read as the epistemic $\mathbb{Z}/3$	$\{J_x, J_y, J_z\}$ <i>commute</i> : a co-measurable triple, not three complementary bases — same count, wrong kind	CLOSED NEGATIVE FTD-0228
The Bell wall — a substrate-native CHSH violation > 2	A local deterministic substrate obeys $S \leq 2$ (Bell’s theorem applies); $S = 2\sqrt{2}$ appears only <i>after</i> the complexification $J_x + iJ_y \mapsto \psi$, which <i>is</i> an instance of M	CLOSED NEGATIVE FTD-0243 §4

The four closures share a moral: where a candidate for non-commutativity seems to appear, it is always either a co-measurable structure wearing the wrong name, or it is the externally imported measurement map M smuggled in by a complexification step. The substrate itself stays commutative throughout.

2.4. FC-1 — the framework declines the measurement map M

FTD-0243 establishes more than the commutativity result: it establishes that non-commutativity is *logically independent* of P1–P5. Both $\{P1 \dots P5\} \cup \{M\}$ and $\{P1 \dots P5\} \cup \{\neg M\}$ are consistent, where M is an independent measurement / quantization postulate (a complexification and projection-valued measure, or a ’t Hooft template-state basis). The theorem proves the *fork exists*. A theorem cannot choose which branch of a genuine fork is true; only a commitment can.

FC-1 (FRAMEWORK COMMITMENT)
The commutative observable algebra A_5 is <i>complete</i> : FTD declines the measurement-map import M . Non-commutativity, Hilbert-space state vectors, and operator-valued observables are not part of FTD’s model of the world. Where quantum mechanics’ non-commutative formalism makes structural predictions that differ from the commutative substrate’s, FTD predicts the substrate.

AXIOM (FTD-0255, an **AXIOM**-class declaration; licensed but not entailed by FTD-0243). **This is a declaration, not a derivation.** The theorem proves the fork; FC-1 picks the $\neg M$ branch. Three consequences follow, each taken up later in the chain: measurement is re-located to the observer layer as a frame-relative epistemic restriction (Chapter 4); the wave-face of quantum phenomenology is carried by the classical flux (Chapter 5); and the structural *differences* between the commutative substrate and the quantum formalism become FTD’s falsifiable spine (Chapter 6). FC-1 places FTD in the ψ -epistemic tradition (a classical ontic substrate with epistemic restrictions) while going beyond it: epistricted theories aim to *reproduce* a fragment of QM, whereas FTD declines the reproduction target itself. What would overturn FC-1 is stated in §2.8.

2.5. FC-2 — the arrow of time is native

The same logical base supports a second declaration. FTD-0253 maps the gap: reversibility is absent from P1–P5, and the genuinely *finite* sector of the substrate—the ternary state field—is irreversible, since manifestation is many-to-one and provably unrecoverable. The reversible part of the dynamics is the continuous flux, on which the finiteness commitment has no purchase. Finiteness therefore actively *opposes* a global reversibility postulate rather than merely failing to imply one.

FC-2 (FRAMEWORK COMMITMENT, ARROW HALF)

The arrow of time is native: the manifestation/state sector is fundamentally irreversible, and FTD declines global reversibility as a postulate. Reversibility holds only as a *sector* property of the weakly-coupled flux wave dynamics—and with it everything that rides on reversibility, including the Lorentzian metric.

AXIOM (FTD-0256, an **AXIOM**-class declaration; licensed by FTD-0253 and the scoped IR measurement FTD-0252). Again **a declaration, not a derivation**: FTD-0253 proves only that the postulates do not *force* reversibility, and that FTD’s finiteness argues against it; FC-2 commits to the irreversible reading as the framework’s official model. Two scope notes keep this honest. First, the causal cone $c = \frac{1}{\sqrt{3}}$ remains forced (**THEOREM**, from P4); it is the metric structure layered on top of the cone—and the Lorentzian space–time mixing it carries—that is declared emergent and sector-scoped, not the cone itself. Second, space (P1) and time (P2) are fundamentally separate primitives at the base: substrate simultaneity is absolute, and Minkowski mixing is an infrared property of the flux wave sector only. The metric half of FC-2 is developed where it belongs, in Chapter 5; here Logic records only the arrow half and its declaration status. What would overturn FC-2 is stated in §2.8.

2.6. The epistemic horizon, honestly split

If the substrate’s logic is classical, why does observing it feel restricted—why can an internal observer not simply read the whole configuration? FTD’s answer splits cleanly into a part it can prove and a part it cannot, and the honesty of the split matters more than either half.

The binding half. A finite internal observer—a subsystem with M pointer states attempting to model a total of $N > M$ states, *itself included*—necessarily loses access to part of its own state. This is ordinary classical finite self-reference (a pigeonhole argument); it uses no quantum input and is non-circular. Every internal frame therefore carries a **self-blind-spot**.

THEOREM BINDING HALF (FTD-0227).

The sharp half. The *Spekkens knowledge-balance*—the blind-spot rotating symmetrically over complementary bases, which is what would yield genuinely quantum-like epistemic statistics—requires the full symplectic $S_3 = \mathbb{Z}/2 \rtimes \mathbb{Z}/3$ action. The $\mathbb{Z}/2$ is FTD-native (the $J^2 = -I$ quarter-conjugacy); the $\mathbb{Z}/3$ is *missing* (the $N_c = 3$ candidate was the same co-measurable triple closed as route 3 above). Binding-without-sharpness is the framework’s honest state.

OPEN (FTD-0227).

A post-hoc note, flagged as such: binding-without-sharpness *after the fact* accounts for the engine’s measured detection statistics being Rice / Gaussian rather than Born (the PL-1 row of Chapter 6)—a classical restriction without the symmetric balance gives classical-with-noise statistics, not $|\psi|^2$. This account was constructed after the measurement; it is an explanation candidate, not a pre-registered prediction, and it is tagged accordingly.

2.7. Reproduced derivation: commutativity independence

This is the load-bearing argument of the layer. It is reproduced in full because the entire weight of FC-1 rests on it, and because its crucial step—that a non-trivial symplectic structure and a commutative observable algebra coexist without contradiction—is exactly the point an outside reader is most likely to doubt. The argument has two claims (absence, then independence), with the Poisson-versus-commutator distinction sitting between them.

Claim A — absence: A_5 is strictly commutative

Every generator in the inventory $\{s, J_a, \text{wave_vel}, \mathcal{L}\}$ is a real-valued function on the single configuration space Ω (an assignment of $(s(v), J(v), \text{wave_vel}(v), \mathcal{L}(v))$ to every voxel v). The algebra’s multiplication is the *pointwise* product, and real multiplication is trivially commutative: for all $\omega \in \Omega$,

$$(A \cdot B)(\omega) = A(\omega) B(\omega) = B(\omega) A(\omega) = (B \cdot A)(\omega).$$

The closure operations preserve function-hood: a Moore-neighbourhood sum of functions is a function, and a composition $A \circ U$ with the deterministic update map is again a function $\Omega \rightarrow \mathbb{R}$. Hence every element of A_5 is a real-valued function on Ω , and the product of any two commutes pointwise. Therefore A_5 is a commutative real algebra. (The algebraic core is machine-checked: `observable_commutator_zero` holds for all pointwise configurations in `lean/Standalone.lean`.) □

The crucial distinction: $\{q, p\} \neq 0$ yet $[q, p] = 0$

Here is the step that defuses the natural objection. The leapfrog / symplectic time-update of the substrate endows its phase space with a *non-zero Poisson bracket*,

$$\{q, p\} = 1,$$

where $(q, p) = (J, \text{wave_vel})$ are the substrate's quadratures. The Poisson bracket is an antisymmetric bilinear *derivation on phase-space functions*. It is a real structure on the *space of beables*—it is what makes the quadrature phase wind, and it is genuinely non-trivial. It is *not* the algebra's multiplication.

The observable commutator is a different object entirely. It is built from the *pointwise product*, and by Claim A that product is symmetric, so

$$[q, p] = q \cdot p - p \cdot q = 0$$

identically, on every configuration. The two facts

$$\boxed{\{q, p\} \neq 0 \quad \text{yet} \quad [q, p] = 0} \tag{2.1}$$

are therefore perfectly consistent: $\{\cdot, \cdot\}$ acts on functions, $[\cdot, \cdot]$ acts on the algebra's product, and a structure on the first does not imply a structure on the second. **THEOREM** (FTD-0243).

The intuition worth carrying away: the substrate is *symplectically rich* and *algebraically commutative* simultaneously. Its quadratures wind—there is a genuine clock—but all quadratures *co-measure*; choosing one as “the” measured value, and declaring the conjugate one disturbed, is not a fact the substrate provides. What would convert the non-trivial Poisson structure into a non-commutative operator algebra is a *deformation quantization*, a map $\{\cdot, \cdot\} \mapsto [\cdot, \cdot]/i\hbar$. That map intrinsically requires the external deformation parameter $\hbar$ —it acts as precisely the external measurement map M —and is not derivable from A_5 alone. The symplectic structure invites quantization; it does not perform it.

Claim B — independence: M cannot be distilled from A_5

By Claim A, A_5 is strictly commutative, so its event lattice is distributive (Boolean) and a joint probability distribution always exists. Any standard QFT or relativistic-time target strictly requires a non-commuting pair, $[A, B] \neq 0$ (a non-distributive event lattice / type III₁ modular algebra). Commutativity, however, is algebraically *preserved* under topological limits and classical coarse-graining: a limit of commuting families commutes, and coarse-graining a Boolean lattice yields a Boolean lattice. Therefore no limit and no coarse-graining applied to A_5 can ever organically yield $[A, B] \neq 0$. The non-commutative structure is logically independent of P1–P5 and must be supplied by an external map M . That $\{P1 \dots P5\} \cup \{M\}$ is consistent is witnessed by the existing emergent-QM constructions that explicitly introduce such an M ; consistency together with non-derivability is exactly logical independence. $\square$

Why the Bell violation does not contradict this

A consistency check, because the framework does exhibit a Bell violation. The $S = 2\sqrt{2}$ value in FTD is *emergent*, derived via the Gauss constraint as a transverse non-local bounding mechanism: the CHSH-violating correlations arise *geometrically* from that constraint, not from native non-commuting measurement operators. But translating those emergent classical correlations into the standard QM Bell format relies on a complexification $J_x + iJ_y \mapsto \psi$ with coarse-graining—and that complexification *is* an instance of the independent map M . The substrate itself remains strictly commutative and limited to $S \leq 2$; the apparent $2\sqrt{2}$ emerges only once M is applied. The theorem holds, sharpening the boundary without contradicting the emergent result. (The unresolved laboratory burden this leaves is PL-2 in Chapter 6, the framework’s highest-risk row.)

What the argument licenses

Claims A and B together are the licensing theorem for FC-1: they prove the fork is genuine (both branches consistent) and that the substrate cannot cross it on its own. They do *not* pick a branch—that is FC-1’s job, and FC-1 is a declaration. The same argument supplies the parallel that organizes the rest of the chain: the relativistic *metric* needs reversibility (declined by FC-2) and the quantum *structure* needs non-commutativity (declined by FC-1), and these are *the same shape of gap*—each a single added principle on the same classical, commutative, irreversible base.

2.8. What this layer does not settle

Logic’s two strongest moves are *declarations*, and the discipline of the chain is to say so and to state in advance what would overturn each. None of the following is softened; the kill-conditions are quoted from the constitution’s falsification register.

FC-1 and FC-2 are declarations, not theorems. The licensing results (FTD-0243 for FC-1, FTD-0253 for FC-2) prove only that the postulates leave the relevant fork open. Picking the branch is a model-defining choice, tagged AXIOM-class and never to be read as derived. A reader who accepts the postulates is *not* thereby compelled to accept either commitment; that is the honest cost of taking them.

What kills FC-1. Any one of: (i) a substrate-native pair of observables with $[A, B] \neq 0$ derived from P1–P5 (this would refute FTD-0243’s premise and is the single most direct refutation of the framework); (ii) a P1–P5 derivation of Born statistics; (iii) a substrate-native, M -free correlation experiment exceeding $S = 2$ beyond numerical error.

What kills FC-2. Any one of: (i) exact γ (exact Lorentz invariance) at finite L off the infrared limit—a measured *absence* of the predicted ultraviolet corrections where the lattice requires them; (ii) a P1–P5 derivation of global reversibility (this would refute FTD-0253’s boundary); (iii) failure of the measured IR approach—the L^{-2} law breaking on the $\langle 100 \rangle$ axis at larger L , or the k^4 anisotropy decay reversing. The detailed kill-conditions and their protocols are the subject of Chapter 6.

The sharpness of the blind spot is open. The binding half of the epistemic horizon is a theorem; the *sharp* half—the symmetric knowledge-balance that would make the observer’s restriction genuinely quantum-like—is unresolved, blocked on the missing $\mathbb{Z}/3$ of the symplectic budget symmetry. OPEN (FTD-0227). Binding-without-sharpness is where the observer story honestly stands; it is carried forward, not papered over, into the philosophy layer (Chapter 4).

3. Mathematics — The Algebraic Spine

The move this layer makes: a forced chain, then a precisely located wall

The ontology of Chapter 1 hands mathematics one modelling choice and a fistful of postulates; this chapter shows what those force. The striking thing is how far the forcing reaches and how sharply it stops. From a single reading of the cubic lattice’s quarter-turn symmetry as the arithmetic of the Gaussian integers $\mathbb{Z}[i]$, one integer is forced ($4 = |\mathbb{Z}[i]^\times|$), then one bridge constant ($G^* = \Gamma(1/4)/\Gamma(3/4)$, reachable four independent ways), then one quadratic ($P(x) = x^2 - 16 G^{*2} x + 16 G^{*3}$) whose larger root sits 1.26 parts per million from $1/\alpha$. Each step is an ordinary, checkable piece of mathematics built on the last.

And then the chain hits a wall — not a failure of effort but a provable boundary. To read the quadratic’s root *as* the physical coupling, the substrate would have to assemble a particular 2×2 operator. The ontology forces the *menu* of that operator’s ingredients but not the *assembly* itself. That gap is the whole honesty of the project: it is why $x_+ = 1/\alpha$ stays a strongly-motivated conjecture and not a theorem, and it is the hinge between the forced mathematics of this chapter and the *matched* physics of Chapter 5.

The reader’s takeaway, fixed in advance, is the canonical accounting of the spine: **nine numbered results, of which seven are theorem-grade and two are honestly tiered below theorem grade**. This chapter states each at its canonical tag, reproduces three load-bearing derivations in full, and maps the boundary precisely. It introduces no new mathematics. Where it disagrees with SPEC_ALGEBRAIC_SPINE.md, the LEDGER, or the construction monograph, those documents are correct and this one has a bug.

3.1. The seed: $\mathbb{Z}[i]$ and the forced 4

The construction begins with one structural choice and one group-theory fact, and the honesty of everything downstream depends on not confusing them.

The cubic lattice’s coordinate planes carry an order-4 rotational symmetry: a 90° rotation of the (x, y) plane cycles $(1, 0) \mapsto (0, 1) \mapsto (-1, 0) \mapsto (0, -1) \mapsto (1, 0)$. FTD’s seed move is to *read this planar quarter-turn as the arithmetic of the Gaussian integers* $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$, with the rotation realized as multiplication by i . That the lattice’s quarter-turn *should be read through* $\mathbb{Z}[i]$ rather than some other order-4 structure is a framework commitment — the discrete-ontology analogue of building $\mathbb{C}$ as $\mathbb{R}[i]$: natural, motivated, and a *choice*. It is the framework commitment FC-0 **AXIOM** (FTD-0254, canonized from FTD-0249); no later theorem may silently inherit the status “forced” from this step, which was “chosen.”

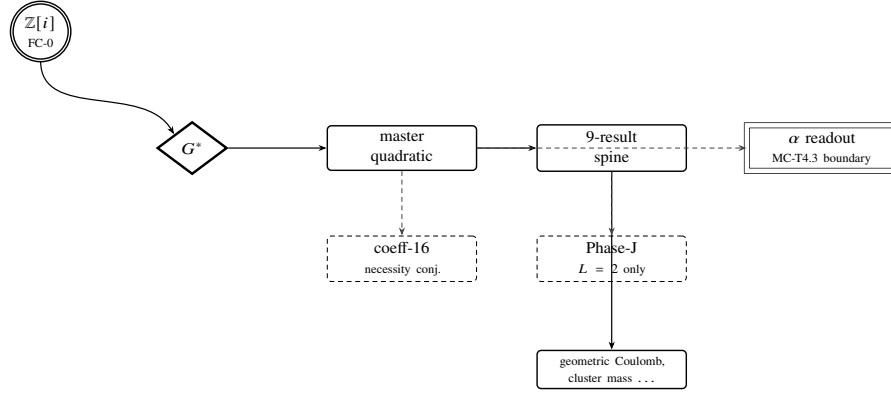


Figure 3.1.: This chapter in the chain. The construction trunk $\mathbb{Z}[i] \rightarrow G^* \rightarrow$ master quadratic $\rightarrow$ the nine-result spine runs in solid (constructed-from) arrows; the dashed links mark the two honestly-tiered results (coefficient-16, Phase-J) and the α -readout boundary, the double-ruled wall where the forcing stops.

Once $\mathbb{Z}[i]$ is the substrate's arithmetic, the rest is ordinary algebra. The unit group is

$$\mathbb{Z}[i]^\times = \{1, i, -1, -i\} \cong \mathbb{Z}/4, \quad |\mathbb{Z}[i]^\times| = 4,$$

since $a + bi$ is a unit iff its norm $a^2 + b^2 = 1$, whose only integer solutions are $(\pm 1, 0)$ and $(0, \pm 1)$ **THEOREM**. The single integer this seed deposits into the construction is $4 = |\mathbb{Z}[i]^\times|$. It reappears, squared, as the coefficient 16 of the master quadratic (§3.3). The discipline to hold: 4 is a *theorem* about $\mathbb{Z}[i]$; that the lattice is entitled to $\mathbb{Z}[i]$ in the first place is the **AXIOM**.

3.2. G^* : the bridge constant, four ways — and why it is not ϖ

The constant at the centre of the spine is

$$G^* = \frac{\Gamma(1/4)}{\Gamma(3/4)} = 2.95867511919\dots \text{ **THEOREM**}$$

(Theorem 1; FTD-0001; OT-1.2). What earns G^* its place is not any one definition but a *convergence*: four constructions of differing character — a Γ -ratio, a lattice Green's-function period, a regularized determinant ratio, and a finite product — all land on the same real number. The four routes, each **THEOREM** (conditional on the cited classical result where one is named):

1. **The Γ -ratio.** Directly from the Euler reflection formula (reproduced in full in §3.6); the canonical statement of OT-1.2.
2. **The Watson BCC period.** The body-centred-cubic Watson integral satisfies $W_3 = \frac{G^{*2}}{2\pi}$ (Theorem 5; OT-2.1), conditional on Watson's 1939 closed-form evaluation, consolidated by Glasser–Zucker (1980). This is G^* arising from a *lattice* quantity, of a wholly different character from the Γ -ratio.

3. **The $\det_\zeta$ quarter-conjugacy bridge.** With J the conjugacy operator ($J^2 = -I$; the same $\mathbb{Z}/4$ seed), the quarter-twisted spectra $D_{1/4}$, $D_{3/4}$ and Lerch's determinant formula $\det_\zeta \{n + a\}_{n \geq 0} = \frac{\sqrt{2\pi}}{\Gamma(a)}$ give the ratio $\frac{\det_\zeta D_{3/4}}{\det_\zeta D_{1/4}} = \frac{\sqrt{2\pi}/\Gamma(3/4)}{\sqrt{2\pi}/\Gamma(1/4)} = \frac{\Gamma(1/4)}{\Gamma(3/4)} = G^*$ (OT-1.7, FTD-0141). The arithmetic content ties it to the seed: $4 \cdot D_{1/4}$ and $4 \cdot D_{3/4}$ are the two non-trivial residue classes mod 4 — the split and inert prime classes of $\mathbb{Z}[i]$.
4. **The finite- N attractor.** The finite product $G_N^* := (N+1)^{-1/2} \prod_{n=0}^N \frac{n+3/4}{n+1/4}$ is computable in $O(N)$ rational operations, and $G_N^* = G^* + \frac{C}{N^2} + O(N^{-3})$, so $G_N^* \rightarrow G^*$ at rate $O(1/N^2)$ (OT-1.8, FTD-0142). This route discharges the undefined-boundary ontology's ε - L obligation for G^* : it supplies a convergent *finite* computation, no completed infinity required.

THE $G^* \neq \varpi$ WARNING (FTD-0117)

$G^* \approx 2.95868$ is the lemniscatic *ratio* $\Gamma(1/4)/\Gamma(3/4)$. It is **not** the Bernoulli/Gauss lemniscate constant $\varpi \approx 2.62206$, a distinct number related by $G^* = \frac{\Gamma(1/4)^2}{\pi\sqrt{2}} = \frac{2\varpi}{\sqrt{\pi}}$ (equivalently $\varpi = \frac{\Gamma(1/4)^2}{2\sqrt{2}\pi} \approx 2.62205755$). The two are sometimes both called “the lemniscate constant” in informal usage; they must never be conflated here. The distinction is load-bearing in the most literal sense: the master quadratic of §3.3 produces $x_+ = 137.0361714582 \dots$ *only* at $G^* = 2.95868$. Substituting $\varpi = 2.62206$ into the same polynomial gives $x_+ \approx 107.3$, nowhere near $1/\alpha$. A 2026-04-30 audit (FTD-0117) found and corrected exactly this conflation across five documents; the canonical value to cross-check against is `G_STAR` in `scripts/constants.py`.

3.3. The master quadratic

With G^* in hand, the construction's central polynomial is purely algebraic. Define

$$P(x) = x^2 - 16 G^{*2} x + 16 G^{*3}. \quad \text{THEOREM}$$

Its discriminant factors cleanly,

$$\begin{aligned} \Delta &= (16 G^{*2})^2 - 4 \cdot 16 G^{*3} \\ &= 256 G^{*4} - 64 G^{*3} = 64 G^{*3} (4G^* - 1), \end{aligned}$$

which is *positive* because $G^* > 1/4$ (indeed $G^* \approx 2.95868$), so both roots are real:

$$\begin{aligned} x_{\pm} &= 8 G^{*2} \pm \sqrt{64 G^{*4} - 16 G^{*3}} \\ &= 8 G^{*2} \pm 4 G^* \sqrt{4 G^{*2} - G^*}. \end{aligned}$$

Numerically $x_+ = 137.0361714582 \dots$ and $x_- = 3.0239639163 \dots$. This is the quadratic formula applied to a polynomial whose coefficients are integer multiples of powers of G^* ; nothing more is involved, and the whole of it is **THEOREM** (Theorem 2; FTD-0001; OT-1.1). The full derivation is reproduced in §3.7.

The coefficient-16 soft spot, flagged honestly. The prefactor 16 has a tempting arithmetic reading. For the lemniscatic CM elliptic curve $E : y^2 = x^3 - x$, the automorphism group over $\mathbb{Q}$ is $\text{Aut}(E) = \{\pm 1, \pm i\} \cong \mathbb{Z}/4$ — the unit group of $\mathbb{Z}[i]$ again — so $|\text{Aut}(E)| = 4$ and $|\text{Aut}(E)|^2 = 16$. That $|\text{Aut}(E)|^2 = 16$ is a **THEOREM** (Theorem 4 arithmetic fact; FTD-0006). But the claim that the master quadratic’s prefactor is *forced* to equal $|\text{Aut}(E)|^2$ — that two distinct objects both equal 16 for a structural reason — is **not proven**; the structural necessity is **CONJECTURE** (OT-4.1, the lone Tier-4 entry: “two distinct objects both equal 16; the structural reason for the coincidence is conjectured but not proved”). This is the framework’s softest mathematical spot. A partial unification exists (the tower-level reading of FTD-0122 supplies a reason for $k = 4$, hence $2^k = 16$, via $|\mathbb{Z}[i]^\times|^2 = 16$), but it is a partial unification, not a forcing theorem.

No physics here, and the retired smaller root. In this chapter, $x_+ = 137.0361714582\dots$ is a root of a polynomial — nothing more. Its identification with $1/\alpha$ is a **STRONGLY MOTIVATED CONJECTURE** (FTD-0013), deferred in full to Chapter 5; it is not asserted as forced anywhere in the construction. The smaller root $x_- \approx 3.024$ is a pure algebraic artifact, fixed by the Vieta relation $x_- = 16 G^{*3}/x_+$ the instant x_+ is fixed, and it carries no physics: the historical identification $x_- \leftrightarrow N_c$ is **retired** (the LEDGER row FTD-0014 was removed in commit ca7eb61). FTD’s $N_c = 3$ is **THEOREM** from topology, independently sourced — the Moore Layer Theorem and the four convergent routes of DERIV_NC_FROM_TOPOLOGY.md — and owes nothing to the polynomial’s smaller root.

3.4. The nine-result spine: seven theorem-grade, two honestly tiered

On the foundation of G^* and the master quadratic, a family of further results stands. The canonical accounting is fixed in SPEC_ALGEBRAIC_SPINE.md §0 and matches the tiering of TRACKER_ONTIC_TRUTH.md: **nine numbered results, of which seven are theorem-grade and two are honestly tiered below theorem grade**. Table 3.1 states each at its exact tag.

Three of the theorem-grade results deserve their statements in line, because they show the spine is a dependency structure and not a list.

Harmonic-invariant tower **THEOREM (Theorem 8, FTD-0111).** For each integer $k \geq 3$, the $(1+i)$ -tower master quadratic is $M_k(x) = x^2 - 2^k G^{*k-2} x + 2^k G^{*k-1}$ (the $k = 4$ instance is the master quadratic of §3.3). Writing the normalized roots $y_\pm := x_\pm/G^*$, then at *every* level $k \geq 3$, $\frac{1}{y_+} + \frac{1}{y_-} = 1$. The proof is a three-line Vieta computation. Furthermore the discriminant factors as $\text{disc}(M_k) = 2^{k+2} G^{*k-1} A_k$, $A_k := 2^{k-2} G^{*k-3} - 1$, with A_k rational at $k = 3$ and *transcendental over $\mathbb{Q}$* at every $k \geq 4$ (via Schneider–Chudnovsky transcendence of G^*).

$\mathbb{Q}(G^*)$ is π -free **THEOREM, conditional on Chudnovsky 1976 (Theorem 9, FTD-0112).** The field generated by G^* shares no algebraic content with $\mathbb{Q}(\pi)$ beyond the rationals: $\mathbb{Q}(G^*) \cap \mathbb{Q}(\pi) = \mathbb{Q}$. The proof reduces a hypothetical polynomial relation between G^* and π — via the identity $G^* \cdot \pi \sqrt{2} = \Gamma(1/4)^2$ — to a relation between π and $\Gamma(1/4)$, which Chudnovsky’s 1976

Table 3.1.: The nine-result algebraic spine, each at its canonical tag. Seven are theorem-grade (results 1, 2, 3, 5, 6, 8, 9); two are honestly tiered below (result 4, coefficient-16 necessity; result 7, Phase-J at $L = 2$ only).

#	Result	Tag	Provenance
1	$G^* = \Gamma(1/4)/\Gamma(3/4)$ algebraic identity (Chowla–Selberg)	THEOREM	FTD-0001
2	Master quadratic $P(x)$ and its two real roots	THEOREM	FTD-0001
3	CM-curve uniqueness within class-number-1 fields (trivial-multiplier criterion)	THEOREM	FTD-0003
4	Coefficient $16 = \text{Aut}(E) ^2$: value-level identity THEOREM ; structural <i>necessity</i>	CONJECTURE	FTD-0006 (T4)
5	Watson identity $W_3 = \frac{G^{*2}}{2\pi}$ on the BCC sub-lattice	THEOREM	FTD-0001 (sub)
6	Phase-G geometric Coulomb $\alpha_r(r, L) = 2r G_L(r)$ (lattice Poisson kernel)	THEOREM	FTD-0004
7	Phase-J partition-function ultralocality, THEOREM AT $L = 2$; disconfirmed for general L	THEOREM	FTD-0005 (T3)
8	Harmonic-invariant tower $\frac{1}{y_+} + \frac{1}{y_-} = 1$ ($A_k \notin \mathbb{Q}$, $k \geq 4$)	THEOREM	FTD-0111
9	Field-theoretic $\mathbb{Q}(G^*)$: maximal π -free subfield (cond. Chudnovsky 1976)	THEOREM	FTD-0112

algebraic-independence theorem forbids. “Conditional” here means “rigour equals this published, foundational transcendence result,” not “depends on a conjecture.”

The two honestly-tiered results, stated plainly. Result 4 (coefficient-16) is the value-level identity above, true as a THEOREM but with structural necessity only CONJECTURE (OT-4.1, Tier 4). Result 7 (Phase-J partition-function ultralocality) is THEOREM AT $L = 2$ only: on a 2^3 lattice the classical Euclidean action depends on the state field solely through $\sum_i s_i^2$ (the manifested-site count) and is permutation-invariant, proved by explicit construction; the general- L extension is *disconfirmed* — the $L = 2$ ultralocality is a Nyquist-mode counting degeneracy (every non-zero momentum has $\sin(k_i) = 0$), not a structural property, and at $L \geq 3$ the action depends on spatial placement (OT-3.1, Tier 3). Both are stated here below theorem grade because that is where their canonical sources place them.

3.5. The α boundary: dynamical, not structural

The spine forces G^* , the coefficient 16, and the polynomial's roots. To turn the polynomial *toward physics* one more thing is needed, and naming it sharply is the entire content of this section. The argument structure is summarized here; the exhaustive four-route case analysis is in the construction monograph (Part II), cited in full below.

The readout move. Read the quadratic as the characteristic polynomial of a 2×2 *readout operator* W whose dominant eigenvalue is to be the measured inverse coupling. For its characteristic polynomial to be P , the operator must have the *operator assembly*

$$(\text{Tr } W, \det W) = (16 G^{*2}, 16 G^{*3}).$$

The question that decides whether α is derived or merely matched is exactly: does $\mathfrak{F}$ — the five postulates, the algebraic spine, and the O_h representation theory of the BCC corner module — *force* this assembly, or is it an imposed selection? This is the central foundational obstruction the canonical sources call **MC-T4.3**.

What is forced: the menu. The canonical attack (FTD-0242) ran four FTD-native routes of differing character (a J -twisted $\det_\zeta$ operator, a BCC body-diagonal response operator, the lemniscatic-CM arithmetic of $E : y^2 = x^3 - x$, and a forced-variational / period-ring route). Every route agrees, with nothing smuggled, on what *is* forward-forced **DERIVED**:

- the **even trace** $16 G^{*2}$ — where $16 = |\mu_4|^2$ for $E : y^2 = x^3 - x$ (FTD-0006) and $G^{*2} = 2\pi \cdot G_{\text{BCC}}(0)$, the Watson BCC self-energy (FTD-0001);
- the **existence of a clean FTD-native odd source**: the J -twisted determinant ratio $\det_\zeta(D_{3/4})/\det_\zeta(D_{1/4}) = \Gamma(1/4)/\Gamma(3/4) = G^*$ (degree 1; the $\sqrt{2\pi}$ cancels). This is strictly stronger than trace-only: it makes $16 G^{*3} = 16 G^{*2} \cdot G^*$ *assemblable*.

What is not forced: the assembly W . What no route forces **FOUNDATIONAL OBSTRUCTION NO-GO** is the *operator assembly itself* — that the same 2×2 readout carrying $\text{Tr } W = 16 G^{*2}$ also has $\det W = 16 G^{*3}$. For a 2×2 operator, trace and determinant are **independent invariants**; fixing the trace leaves the determinant free. The odd scalar G^* exists, but nothing forces the *gluing* — neither that the two invariants belong to one operator, nor that the odd scalar lands in the determinant slot. The verdict across all four routes is therefore **0 of 4 force the assembly**, and (by the route-invariance theorem below) this is one structural fact reached four ways, not four independent failures. The canonical phrasing: **α is dynamical, not structural** — the ontology forces the *menu* but not the *dish*. This is a **FOUNDATIONAL OBSTRUCTION NO-GO**, **not a theorem and not a derivation of α** .

The boundary is built from theorem-grade masonry. The deeper attack (FTD-0243) established three genuinely theorem-grade facts that hem the open kernel into a single, precisely located gap:

1. **The flip is ruled out** THEOREM. The only geometric candidate for an operator that forces $\det = 16 G^{*3}$ — the $C_3(\langle 111 \rangle)$ three-plane $\det_\zeta$ product — is excluded from the rank-2 readout: the complex structure J ($J^2 = -I$) required by the trace forces O_h to break to one C_4 axis, so $C_3(\langle 111 \rangle) \notin \text{Stab}$, while the three-plane product requires it; and $\langle C_4, C_3 \rangle = O$ (machine-checked) leaves no readout.
2. **Leg 3b closes its own scope** THEOREM. No $C_3(\langle 111 \rangle)$ -equivariant rank-2 restriction of the three-plane source carries $(\text{Tr}, \det) = (16 G^{*2}, 16 G^{*3})$; the chain is reality $\Rightarrow$ scalar- $i \Rightarrow C_4 \Rightarrow O$ — the same wall reached from the restriction side.
3. **The reduction is route-invariant** THEOREM. All four attacks reduce to the same obligation because $\mathbb{Q}(G^*)$ is the Galois-fixed field of the master quadratic's $\mathbb{Z}/2$: the splitting field over $\mathbb{Q}(G^*)$ is $\mathbb{Q}(G^*)(\sqrt{G^*(4G^* - 1)})$, and every forward-forced *symmetric* datum (the Watson trace, the $\det_\zeta$ ratio, the Chowla–Selberg periods) lives in the σ -fixed subfield $\mathbb{Q}(G^*)$ and is provably blind to which root is $1/\alpha$.

The square-root wall and the conditional theorem. In the field dialect, the residual obligation is the cleanest form and gives the boundary its name (**K-BIND**): to fix α , the substrate must *natively realize* the squarefree generator $\sqrt{G^*(4G^* - 1)}$ of the quadratic extension $\mathbb{Q}(G^*)(\sqrt{G^*(4G^* - 1)})/\mathbb{Q}(G^*)$ — the one antisymmetric beable that breaks the $\mathbb{Z}/2$ root-swapping symmetry and selects the physical branch. Every symmetric datum cannot tell the two roots apart; this square root is exactly the missing ingredient. K-BIND is CLOSED-NEGATIVE (FTD-0244): axiomatizing the admissible operator calculus over $V_{\text{complex}} \cong \mathbb{Z}[i]^\times$ -module $\mathbb{Z}[i]^2$ forces all invariants into $\mathbb{Q}(G^*)$, and the degree-2 extension cannot be natively realized (transcendence of G^* , Chudnovsky 1976), so no native operator reaches x_+ without the external selection W . What *is* provable unconditionally is the conditional theorem THEOREM: $\mathfrak{F}$ does not force α — the assembly $(\text{Tr}, \det) = (16 G^{*2}, 16 G^{*3})$ is logically independent of $\mathfrak{F}$ — unless $\mathfrak{F}$ is extended by a substrate-native binding law W fixing the determinant's odd- G^* exponent at exactly 1. The independence is witnessed on both sides: $\mathfrak{F} \cup \{W\}$ is consistent (the master quadratic is the witness model), and $\mathfrak{F} \cup \{\neg W\}$ is consistent ($\det = 16 G^{*4}$ and $\det = G^*$ are explicit, $\mathfrak{F}$ -consistent alternative assemblies). W is therefore a 6th-postulate-class input (FTD-0243).

Net. MC-T4.3 remains a FOUNDATIONAL OBSTRUCTION NO-GO, and $x_+ = 1/\alpha$ (**FTD-0013**) remains a STRONGLY MOTIVATED CONJECTURE — nothing here promotes either. The contrast with $N_c = 3$ is exact: N_c is structural, forced from O_h and topology by four independent routes with no operator-assembly choice; α has no such forcing. The exhaustive four-route case analysis, with each force-pass and adversarial refute-pass, is the climax (Part II) of the construction monograph (MONOGRAPH_FTD_CONSTRUCTION.md); this section reproduces its argument structure, not its full machinery.

3.6. Reproduced derivation I: the G^* identity

The first load-bearing chain reproduced in full. *Goal:* establish the two standard closed forms of G^* from the definition and the Euler reflection formula alone.

Step 1 — the reflection value. The Γ -function reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$ at $z = 1/4$ gives

$$\Gamma(1/4)\Gamma(3/4) = \frac{\pi}{\sin(\pi/4)} = \pi\sqrt{2}$$

since $\sin(\pi/4) = 1/\sqrt{2}$.

Step 2 — the first closed form. By definition $G^* = \Gamma(1/4)/\Gamma(3/4)$. Multiply numerator and denominator by $\Gamma(1/4)$:

$$G^* = \frac{\Gamma(1/4)}{\Gamma(3/4)} = \frac{\Gamma(1/4)^2}{\Gamma(1/4)\Gamma(3/4)} = \frac{\Gamma(1/4)^2}{\pi\sqrt{2}},$$

substituting the Step-1 value $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$ in the last equality.

Step 3 — the lemniscate form. The Bernoulli/Gauss lemniscate constant is $\varpi = \frac{\Gamma(1/4)^2}{2\sqrt{2}\pi} \approx 2.62205755$. Solving for $\Gamma(1/4)^2 = 2\sqrt{2}\pi \varpi$ and substituting into Step 2,

$$G^* = \frac{2\sqrt{2}\pi \varpi}{\pi\sqrt{2}} = \frac{2\varpi}{\sqrt{\pi}}.$$

Collecting Steps 2–3 gives the canonical closed-form chain

$$G^* = \frac{\Gamma(1/4)^2}{\pi\sqrt{2}} = \frac{2\varpi}{\sqrt{\pi}},$$

numerically $G^* = 2.95867511919 \dots$ THEOREM (Theorem 1; FTD-0001). This is a three-line proof.

Note that the lemniscate form makes the $G^* \neq \varpi$ distinction quantitative: G^* and ϖ differ by the factor $2/\sqrt{\pi} \approx 1.128$, so $G^* \approx 2.95868$ while $\varpi \approx 2.62206$. They are never interchangeable.

3.7. Reproduced derivation II: the master-quadratic assembly

The second chain, reproduced from the (trace, determinant) data to the explicit roots. *Goal:* given the assembly $(\text{Tr } W, \det W) = (16 G^{*2}, 16 G^{*3})$, recover $P(x)$, its discriminant, and $x_{\pm}$ in closed form.

Step 1 — the characteristic polynomial. A 2×2 operator W with trace $\text{Tr } W$ and determinant $\det W$ has characteristic polynomial $x^2 - (\text{Tr } W)x + (\det W)$. Inserting the assembly $(\text{Tr } W, \det W) = (16 G^{*2}, 16 G^{*3})$ gives exactly

$$P(x) = x^2 - 16 G^{*2} x + 16 G^{*3}.$$

Step 2 — the discriminant. For $x^2 - bx + c$ the discriminant is $\Delta = b^2 - 4c$. With $b = 16 G^{*2}$ and $c = 16 G^{*3}$,

$$\begin{aligned} \Delta &= (16 G^{*2})^2 - 4 \cdot 16 G^{*3} \\ &= 256 G^{*4} - 64 G^{*3} = 64 G^{*3} (4G^* - 1), \end{aligned}$$

The factored form $64 G^{*3} (4G^* - 1)$ is manifestly positive because $G^* \approx 2.95868 > 1/4$, so both roots are real and distinct.

Step 3 — the roots. The quadratic formula $x_{\pm} = \frac{1}{2}(b \pm \sqrt{\Delta})$ with $b = 16 G^{*2}$ and $\sqrt{\Delta} = \sqrt{256 G^{*4} - 64 G^{*3}} = 8\sqrt{4 G^{*4} - G^{*3}}$ gives

$$\begin{aligned} x_{\pm} &= 8 G^{*2} \pm \sqrt{64 G^{*4} - 16 G^{*3}} \\ &= 8 G^{*2} \pm 4 G^* \sqrt{4 G^{*2} - G^*}. \end{aligned}$$

the second line pulling $4 G^*$ out of the radical. Numerically $x_+ = 137.0361714582 \dots$ and $x_- = 3.0239639163 \dots$ **THEOREM** (Theorem 2; FTD-0001). Everything here is the quadratic formula on a polynomial with G^* -power coefficients; no physics has entered, and none is claimed.

3.8. Reproduced derivation III: the boundary argument structure

The third reproduction is not a numerical derivation but the *argument structure* of the α boundary — self-contained enough that the reader can see precisely what is forced and what is not, without the full four-route machinery (which is cited to the construction monograph, Part II, FTD-0242).

Premise (the readout move). To read x_+ as the physical inverse coupling, posit a 2×2 readout operator W whose characteristic polynomial is P . By §3.7 Step 1 this is equivalent to requiring the assembly $(\text{Tr } W, \det W) = (16 G^{*2}, 16 G^{*3})$.

The forced half (menu). The even invariant is forward-forced: the trace slot $16 G^{*2}$ is assembled from $16 = |\mu_4|^2$ and the Watson self-energy G^{*2} , with G^* kept symbolic and no banned move. A clean degree-1 odd source also exists: $\det_{\zeta}(D_{3/4})/\det_{\zeta}(D_{1/4}) = G^*$. Hence $16 G^{*3} = 16 G^{*2} \cdot G^*$ is *assemblable in principle*. This half is **DERIVED**.

The unforced half (assembly W). Trace and determinant of a 2×2 operator are independent invariants: knowing $\text{Tr } W = 16 G^{*2}$ places no constraint on $\det W$. So the existence of an odd scalar G^* does not force that *this* operator carries it in *this* slot at *this* exponent. The free family $\det = 16 G^{*2} \cdot G^{*k}$ for $k = 0, 1, 2, 3$ yields dominant roots $139.05/137.04/130.68/105.76$ respectively — all $\mathfrak{F}$ -consistent, and **nothing in $\mathfrak{F}$ selects $k = 1$** . This half is **FOUNDATIONAL OBSTRUCTION NO-GO**.

Why the verdict is route-invariant (not an argument from absence). The four routes are four dialects of one obligation because $\mathbb{Q}(G^*)$ is the Galois-fixed field of the master quadratic's $\mathbb{Z}/2$ (the discriminant's square root $\sqrt{G^*(4G^* - 1)}$ is the squarefree generator of the degree-2 extension $\mathbb{Q}(G^*)(\sqrt{G^*(4G^* - 1)})/\mathbb{Q}(G^*)$). Every *symmetric* forward-forced datum lives in $\mathbb{Q}(G^*)$ and is therefore provably blind to which root is $1/\alpha$. The single ingredient that would break the symmetry — the antisymmetric beable $\sqrt{G^*(4G^* - 1)}$ — is exactly the one the substrate is not shown to produce. That is K-BIND, **CLOSED NEGATIVE** (FTD-0244).

Conclusion. $\mathfrak{F}$ forces the menu and not the dish; the assembly W is logically independent of $\mathfrak{F}$ on present evidence (conditional theorem, **THEOREM**, FTD-0243); therefore α **is dynamical, not structural**, and $x_+ = 1/\alpha$ stays **STRONGLY MOTIVATED CONJECTURE** (FTD-0013). The construction halts at the discrete limit of a square root — a mapped wall, not a failure of effort.

3.9. What this layer does not settle

The spine is firm; what it does *not* settle is named here, at honest tags.

- **The coefficient-16 necessity.** That $|\text{Aut}(E)|^2 = 16$ is a THEOREM; that the master quadratic's prefactor is *forced* to equal it is CONJECTURE (OT-4.1, Tier 4) — the framework's softest mathematical spot. A forcing theorem (beyond the partial $k = 4$ unification of FTD-0122) is OPEN.
- **Phase-J beyond $L = 2$.** Ultralocality is THEOREM AT $L = 2$ only; the general- L extension is disconfirmed (a Nyquist-mode counting degeneracy, not a structural property). There is no general- L ultralocality theorem and none is sought.
- **The α -readout assembly W .** MC-T4.3 stays a FOUNDATIONAL OBSTRUCTION NO-GO; the unconditional “no FTD-native W can ever exist” is the universal negative, with K-BIND CLOSED NEGATIVE (FTD-0244) on the axiomatic side and ARC-D1 CLOSED NEGATIVE on the empirical side (0 macroscopic cluster fissions across 2000 seeds; a rigid lattice). The two live exits — a 6th-postulate-class binding law W that *forces* the assembly, or a fresh engine-native ARC-D measurement — are OPEN, and neither is presently open.
- **The physics identification itself.** $x_+ = 1/\alpha$ is a 1.26-ppm STRONGLY MOTIVATED CONJECTURE (FTD-0013), not a derivation. Its structural-uniqueness evidence and its honest limit belong to Chapter 5, where the matched physics is quarantined at its tags. The hinge this chapter establishes — forced mathematics, unforced assembly — is exactly why that identification cannot stand higher than a strongly-motivated conjecture.

4. Philosophy — The Observer Layer

The move this layer makes: who reads the lattice, and at what cost

Every preceding layer spoke in the third person. Ontology said *what exists*; logic said *what follows*; mathematics said *what is forced*. None of them needed an observer. The substrate evolves under the five postulates with absolute, god’s-eye simultaneity whether or not anything inside it is looking. The move the philosophy layer makes is to ask the first-person question that quantum mechanics built its entire formalism around — *who reads the lattice, and what can a reader embedded in the lattice know?* — and to answer it *without* importing the quantum measurement apparatus to do so.

The intuition is a deliberately deflationary one. An “observer” in FTD is not a mind, a consciousness, or an agent; it is a *reference frame structure* — a subsystem of the lattice that happens to contain a model of itself. The canonical example is mundane: the centre voxel of a 27-block, which “sees” its world only through the surrounding shell. Reading is restriction. What a frame can know of the configuration — including of itself — is whatever survives the projection from the full state down to the frame’s own readout. There is no collapse here, no special role for awareness, and no deformation of the underlying algebra. Measurement becomes epistemics: the bookkeeping of what a finite thing inside a classical world is structurally barred from knowing about itself.

That deflation buys something most interpretations cannot: *exposure*. Because the early layers are committed — the postulates are frozen, the framework commitments are declared — the observer layer cannot quietly absorb whatever the laboratory reports. When a measurement arrives that the quantum formalism reproduces exactly, an interpretation that *also* imports that formalism is never separated from it by data. FTD, having declined the import, *can* be. The thesis of this chapter is that ontology-first is therefore *more* falsifiable, not less — and the supersession matrix is the precise ledger of where FTD stakes outcomes its neighbours decline to stake. This is an argued layer: it carries no derivations of its own, only the disciplined re-statement of claims proved elsewhere and the honest accounting of what is still open.

4.1. The vocabulary: reference frame structure, not consciousness

FTD’s account of observation is stated entirely in the canonical vocabulary of the reference-frame reference document (REF_REFERENCE_FRAME_VOCABULARY.md), which exists precisely to strip the qualia and Hard-Problem baggage from terms the corpus once used loosely. The load-bearing distinction is between a *structural* property and a *dynamical* one:

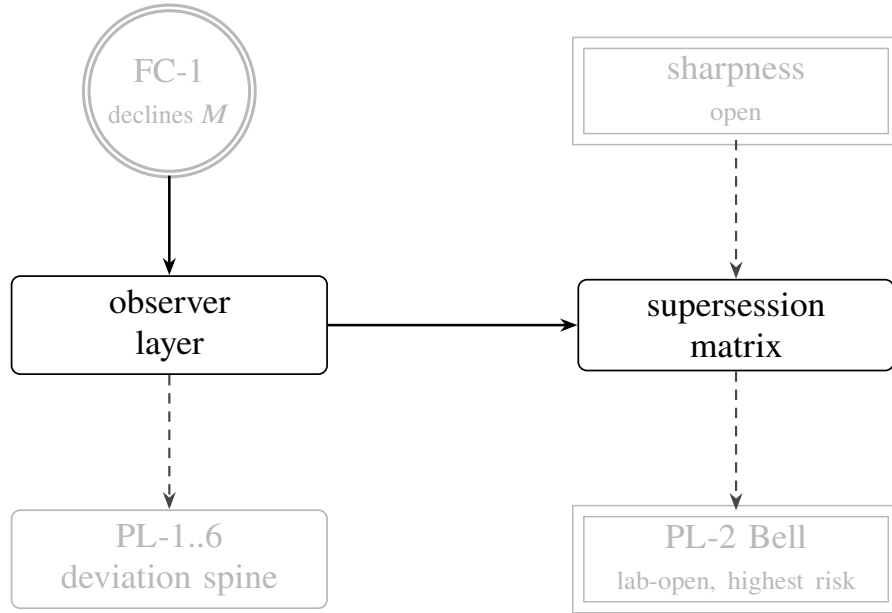


Figure 4.1.: This chapter in the chain. The observer layer and the supersession matrix (solid-thin *selection/synthesis* boxes) sit between the logic layer that fixes what they may claim (FC-1, which declines the measurement-map import) and the science layer they are accountable to (the deviation spine, and the lab-Bell wall that is the framework’s highest-risk row). Ghosted nodes are drawn from adjacent bands for orientation only.

- **Reference frame structure** (structural) — the property of a subsystem that contains a model of itself: a fixed-point / strange-loop property. A *local reference frame* is the lattice region that instantiates it (canonically, the centre of a 27-block). The *reference frame projection* is the operator mapping the full (noumenal 3^3) state to the frame’s (phenomenal 2^3) readout; in algebraic terms it is an observable-algebra *restriction*. The *frame-relative readout* is what the structure thereby knows of itself — the deflated replacement for “experience.”
- **Active frame dynamics** (dynamical) — the capacity of a system to select its own trajectory by its own internal frame-relative readout, as distinct from being driven externally. This replaces the loaded terms “agency” and “free will” with a physics-compatible one, and remains embedded in the deterministic dynamics of P5.

Three discriminations are made explicit, because the framework needs them and the old language hid them. First, reference frame structure is *not* the same as realised active frame dynamics: almost every voxel sits at the centre of some 27-block, so reference frame structure is generic; what is rare is its *realisation* in trajectory selection (the *active-frame threshold* remains an open structural problem). Second, a frame-relative readout is *not* guaranteed faithful — there is no axiom that a frame reads itself correctly, which is exactly the absence-of-omniscience any physical observer must have. Third, active frame dynamics is *not* libertarian free will: the trajectory is selected by frame-internal structure, not by a non-physical chooser. The Hard Problem is not solved here; it is *re-described* as the open problem of frame-relative emergence and then set aside

as out of scope.

4.2. The observer layer: frames, readouts, and the binding blind spot

The layered picture has two strata (FTD-0254 §4.1, [SYNTHESIS] + [OPEN]). The *substrate frame* is the god's-eye view: it holds the full configuration and evolves deterministically under P1–P5 with absolute simultaneity. *Internal frames* are subsystems whose access to the world — and to themselves — is restricted by the reference frame projection. What quantum mechanics treats as the measurement problem, FTD treats as the epistemics of internal frames embedded in a classical substrate: *measurement is frame-relative readout under a derived restriction, not a dynamical collapse and not an algebra deformation* (this last clause is FC-1, carried over from the logic layer).

The one genuinely *derived* ingredient of this picture is the blind spot, and it is honestly split.

- **The binding half** [THEOREM PARTIAL] (FTD-0227). A finite internal observer — a subsystem with M pointer states modelling a total of $N > M$ states, *itself included* — necessarily loses access to part of its own state. This is classical finite self-reference (a pigeonhole / diagonal argument); it is non-circular and takes no quantum input. Every internal frame therefore carries a **self-blind-spot**. The corresponding LEDGER row (FTD-0227) records this binding half as DERIVED and the whole row as PARTIAL; the conservative reading used throughout this document is “[THEOREM], partial.”
- **The sharp half** [OPEN]. The Spekkens *knowledge-balance* — the blind spot rotating symmetrically over complementary bases, the feature that would make the restriction genuinely quantum-like — requires the full symplectic $S_3 = \mathbb{Z}/2 \rtimes \mathbb{Z}/3$ action. The $\mathbb{Z}/2$ is FTD-native (the $J^2 = -I$ quarter-conjugacy); the $\mathbb{Z}/3$ is **missing** (the $N_c = 3$ candidate for it was diagnosed as apophenia, FTD-0228, [CLOSED NEGATIVE]). *Binding-without-sharpness* is the framework's honest state on observation.

A consequence, flagged as a *post-hoc* account rather than a prediction: a classical restriction without the symmetric balance yields classical-with-noise detection statistics rather than Born's $|\psi|^2$ — which is consistent with the engine's measured Rice/Gaussian threshold statistics (§6 PL-1). This explanation was constructed *after* the measurement; it is an explanation candidate, not a pre-registered prediction, and is tagged accordingly.

The honest gaps kept open at this layer are three: the *sharpness* of the restriction just described ([OPEN]); the *active-frame threshold* (which configurations realise active frame dynamics — an open structural problem); and a dedicated formalisation of frame self-identification (the GUID+XYZ construction), which does not yet exist as a foundational document.

4.3. Position among the alternatives — the supersession matrix

FTD's posture toward the standing interpretive and foundational frameworks is *structural, not rhetorical* (FTD-0254 §4.2, [SYNTHESIS]; per-cell tags as marked). Each row of Table 4.1 is a

respectable research tradition. FTD differs from each in two specific ways: it *declines* an *import* the tradition makes (most often the non-commutative Hilbert-space formalism, sometimes the continuum, sometimes reversibility), and in return it *offers deviations* where the tradition reproduces the standard formalism exactly. The matrix is the ledger of those two moves.

Three structural contrasts carry the supersession claim, and each is a re-statement, not a new argument.

1. **Against the QM family (Copenhagen / MWI / RQM / Bohm).** All four import the non-commutative Hilbert-space formalism as fundamental. FTD *proves* its substrate cannot generate that formalism (the observable algebra A_5 is strictly commutative, THEOREM, FTD-0243) and then *declines* it (FC-1, AXIOM-class, FTD-0255). Where the four reproduce textbook QM exactly, FTD predicts *structural deviations* (§6). FTD is thereby *more falsifiable, not more general*: it stakes outcomes the others cannot.
2. **Against 't Hooft's CA program — the nearest relative.** Both posit a deterministic, local, discrete classical substrate. 't Hooft's program *aims to reproduce QM exactly* via a template-basis map (an instance of the import M) and hopes for reversible automata. FTD declines M and, decisively, declines reversibility too — its own finite (manifestation) sector is provably irreversible and argues *against* a global reversibility postulate (FC-2, AXIOM-class, FTD-0256). Same substrate class; opposite commitments on both forks.
3. **Against string theory.** FTD's structural null-predictions are direct contradictions of string theory's characteristic imports: no SUSY partners, no extra dimensions, no monopoles, exact proton stability (the nulls are THEOREM-grade as a class, §6 PL-6; the proton sub-claim specifically carries a SELECTION correction, FTD-0301). Continued null results at colliders and proton-decay observatories are evidence *for* FTD's ontology and *against* those imports.

4.4. Why ontology-first is more falsifiable, not less

The chain Ontology > Logic > Mathematics > Philosophy > Physics > Science would be vacuous if “Science” merely decorated the front of the chain (FTD-0254 §4.3, SYNTHESIS). The constitution's discipline is the reverse, and it is the central philosophical claim of this layer: *because the early layers are committed, the framework cannot retreat when measurements arrive*.

The argument is short and turns on a single asymmetry. A framework that reproduces the QM/SR formalism exactly — by importing it — can never be *separated* from QM/SR by experiment: any experiment QM/SR survives, the importing framework survives by construction. Its empirical content relative to QM/SR is therefore null. FTD, by *declining* the imports (FC-1, FC-2), forfeits that safety. Each declination converts a borrowed certainty into a stakeable prediction:

- declining the measurement map M (FC-1) turns “the substrate must reproduce Born statistics” into the deviation PL-1 (Rice, not Born), which a single clean construction can refute;
- declining global reversibility and the substrate-level Lorentzian metric (FC-2) turns “ γ is exact at all scales” into the deviation PL-4 (a γ -departure that must die as L^{-2} , and must *not* vanish

where the lattice requires it), and “space is exactly isotropic” into PL-5 (a UV anisotropy dying as k^4);

- committing to the discrete ontology turns string theory’s open-handedness about SUSY, extra dimensions, and proton decay into **THEOREM**-grade nulls (PL-6) that a single positive detection would break.

Crucially, the commitments are themselves staked. The constitution states in advance the observations that kill each one (§6, FC-1 / FC-2 kill-conditions): a substrate-native pair of observables with $[A, B] \neq 0$, or a P1–P5 derivation of Born statistics, would kill FC-1; exact γ at finite L , or a P1–P5 derivation of global reversibility, would kill FC-2. An ontology that names its own executioners is the opposite of an unfalsifiable one.

This is not falsifiability-as-sole-criterion smuggled back in. The framework’s broader stance rejects treating falsifiability as the *only* mark of a good theory; ontology-first is also a commitment to saying what *is* before saying what is *measured*. But rejecting falsifiability-as-sole-criterion is not licence to wave measurements away. The discipline that backs the claim is mechanical: *every* measured phenomenon must appear somewhere — in the IR catalog (§5), in the deviation ledger (§6), or in the not-yet-delivered honesty row of Table 4.1. Nothing measured is permitted to be silent.

4.5. What this layer does not settle

The observer layer is the most interpretively ambitious layer of the chain, and correspondingly the one with the most that is still **OPEN**.

- **The sharpness of the restriction** **OPEN**. The binding blind-spot is a theorem (FTD-0227, partial); the *symmetric* knowledge-balance that would make it quantum-like is not — the missing $\mathbb{Z}/3$ is a genuine gap, not a notational one. Until it closes, the framework’s account of measurement is *binding-without-sharpness*, and this document claims no more.
- **The observer-layer Bell account** **SELECTION** + **OPEN**. The substrate proves $S \leq 2$; laboratories measure $S > 2$. FTD’s account of that gap lives at the observer layer and is a *selection*, not a derivation — and it is the framework’s **highest-risk** commitment. It is carried forward, unsoftened, as PL-2 in §6.
- **The active-frame threshold** **OPEN**. Which configurations cross from mere reference frame *structure* (generic) into realised active frame *dynamics* is an unsolved structural problem; the framework has the vocabulary for it but not the criterion.
- **The Hard Problem** is *not* addressed. It is re-described as the open problem of frame-relative emergence and explicitly placed out of scope. This layer accounts for what a frame can *know* of itself, not for why there is something it is like to be one.

The supersession posture, then, is exactly what it says: a program with a falsifiable spine, holding two named walls open — the sharpness gap and the lab-Bell burden — and accountable, layer by layer, to the physics and the science that follow.

Table 4.1.: **The supersession matrix.** FTD’s position relative to seven standing frameworks, read along the axes that carry the supersession claim. “Non-commutativity” is the quantum measurement formalism each tradition treats as fundamental; FTD *declines* it (FC-1; the independence is [THEOREM](#), FTD-0243). The final row of the table body is mandatory and is *not* a framework: it states, in the framework’s own voice, what FTD has *not* delivered.

Framework	Ontic substrate / determinism / locality	Non-commutativity	Metric & arrow
Copenhagen	none specified (ψ not ontic); not deterministic; no sub- ψ dynamics	fundamental (primitive collapse)	imported background; arrow from measurement
Many-Worlds	universal ψ ; deterministic (unitary); no 3-space locality of outcomes	fundamental (branching)	imported background; decoherence arrow
RQM (Rovelli)	relational facts only; not deterministic; relational	fundamental	imported / relational; perspectival arrow
Bohmian	particle config + ψ ; deterministic; explicitly nonlocal	fundamental (via ψ)	imported background; equilibrium arrow
’t Hooft CA	classical cellular automaton; deterministic; local	imported deliberately (the CA $\rightarrow$ QM map)	imported; aims at <i>reversible</i> automata
Spekkens	classical ontic states; deterministic (toy); local (toy)	reproduced <i>fragment</i>	n/a (toy); knowledge-balance postulated
String theory	strings/branes on a target manifold; quantum; QFT-local	fundamental	dynamical but presupposes continuum + extra dimensions
FTD	ternary lattice + flux (P1–P5, FC-0); deterministic (P5); dynamically local (P4)	declined (FC-1; independence THEOREM , FTD-0243)	cone forced THEOREM ; metric emergent-IR & sector-scoped (FC-2); arrow native (FC-2); space $\perp$ time fundamental

What FTD has *not* delivered (mandatory honesty row). FTD does *not* currently derive: the Born rule (FTD-0187, [SELECTION](#) / [OPEN](#)); the tested constructions FTD-0199/FTD-0200 are [CLOSED NEGATIVE](#); *laboratory* Bell violations $S > 2$ (the observer-layer account is [SELECTION](#)-grade — the substrate bound is $\bar{S} \leq 2$ and labs measure $S > 2$, an **accepted open burden**, §6 PL-2); interference / double-slit phenomenology at the observer level; QFT scattering amplitudes; α (**dynamical, not structural** — $x_+ = 1/\alpha$ stays [STRONGLY MOTIVATED CONJECTURE](#), FTD-0013); and the sharp epistemic balance (§4.2, [OPEN](#)). The supersession posture is a *program with a falsifiable spine*, not a completed replacement.

5. Physics — The Computational EFT

The move this layer makes: run the lattice, read off the IR

Up to here the chain has been a construction. The ontology fixed what exists (Chapter 1), the logic fixed what follows and what is declined (Chapter 2), the mathematics forced a single bridge constant and a single quadratic and then ran into a wall (Chapter 3), and the philosophy asked who reads the lattice (Chapter 4). This chapter asks the empirical question: *when you actually run the substrate, what infrared physics comes out?*

The answer has two registers, and keeping them apart is the whole honesty of the project. In the first register the lattice, simulated as an instrument, *reproduces* a long list of infrared behaviour — a static $1/r$ potential, a relativistic causal cone, a mass gap, cluster masses that track particle masses, a gravitational hierarchy of the right order, confinement signatures, Ward identities. Several of these are not fitted; they are forced by lattice geometry and fall out of the dynamics with zero free parameters. In the second register sits the one quantity every outsider asks about first: the fine structure constant α . Here the framework can only *match*, not *force*. The master quadratic of Chapter 3 has a root that agrees with α^{-1} to roughly one part per million — but Chapter 3’s route-invariance boundary (MC-T4.3) showed that no FTD-native route compels the operator assembly behind that quadratic. So the identification stays a strongly-motivated conjecture, and this chapter says so plainly.

The intuition to carry out of this chapter is therefore a discipline, not a triumph: the computational effective field theory *measures and derives* infrared physics through the engine, each result wearing its own epistemic tag; it reproduces a great deal; and at exactly the place where it would be most spectacular to claim a derivation — α from pure lattice geometry — it declines to overclaim. The accounting at the end of the chapter (~23 derived, ~129 parametric, ~10 imposed) is the quantitative form of that discipline.

5.1. The IR catalog: what the engine reproduces

The physics layer of FTD is a *computational effective field theory*: the infrared behaviour of the substrate as measured and derived through the engine as instrument, under the pre-registration discipline of Chapter 6. The constitution (FTD-0254) records the catalog as fifteen rows, each carrying its own source tag; physical readings are conditional on the calibration declared in the front matter. Table 5.1 reproduces the catalog at those tags, verbatim in epistemic status. It is the empirical census of the framework: every measured phenomenon either appears here, in the deviation ledger of Chapter 6, or in the “not yet delivered” honesty row of Chapter 4 — nothing is waved away.

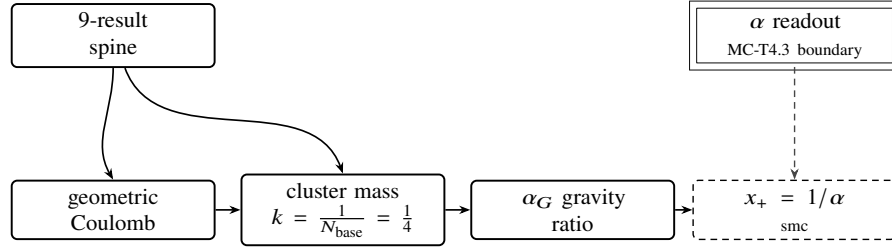


Figure 5.1.: This chapter in the chain. The spine of Chapter 3 discharges into IR physics along solid (*constructed-from*) arrows: geometric Coulomb, the cluster-mass coefficient $k = \frac{1}{N_{\text{base}}} = \frac{1}{4}$, and the gravity ratio α_G are theorem-/derivation-grade. The one number every reader wants, $x_+ = 1/\alpha$, sits in a *dashed* (below-theorem) box: the dashed arrow down from the α -readout wall (the MC-T4.3 boundary of Chapter 3) is exactly why it remains a strongly-motivated conjecture rather than a theorem.

Table 5.1.: The IR catalog v1 (constitution §5.1, FTD-0254): the fifteen engine-native infrared results, each at its source tag. “Engine-native” means the statement is about the simulated substrate; the physical reading is conditional on the calibration. Load-bearing rows are reproduced in full in §5.3.

#	IR result	Tag	Content (engine-native statement; source)
1	Causal cone	THEOREM	$c = \frac{1}{\sqrt{3}}$, forced by Moore locality (P4); cone bit-identical between second- and first-order dynamics (FTD-0253).
2	Geometric Coulomb (Phase G)	THEOREM	$\alpha_r(r, L) = 2r G_L(r)$ — the emergent static potential <i>is</i> the periodic lattice Poisson Green’s function, zero free parameters; $R^2 = 1.0000$ at $L = 384$, 0.07% median tail residual (FTD-0004). Reproduced in §5.3.1.
3	Lorentz / isotropy recovery	MEASURED	the rotation-breaking operator scales as $\delta \propto k^4$ (anisotropy exponent $p = 4.0008 \pm 0.0006$) — strongly irrelevant in the IR.
4	Time dilation / γ -emergence	MEASURED (scoped)	departure from exact γ vanishes as $R \propto L^{-2}$ on $\langle 100 \rangle$ at $v \lesssim 0.85$; ultra-relativistic diagonals unconverged [OPEN] ; clock hypothesis stays AXIOM (FTD-0252).
5	Per-voxel mass gap	THEOREM	the Gauss constraint plus lattice topology force a nonzero per-voxel excitation cost (FTD-0044).
6	Cluster mass = N , coefficient $k = \frac{1}{N_{\text{base}}} = \frac{1}{4}$	DERIVED	$\text{mult}(A_{1g}) = 4$ in the 27-block THEOREM $\Rightarrow$ mean-energy coefficient $\frac{1}{4}$ at linear level; non-linear bridge [OPEN] ; cluster-mass identification [STRONGLY MOTIVATED CONJECTURE] (FTD-0110). Reproduced in §5.3.2.

continued on next page

Table 5.1, continued

#	IR result	Tag	Content (engine-native statement; source)
7	Cluster transport inertia = $N \cdot M_{\text{REST}}$	IMPOSED	the engine honours the action's equivalence principle (demonstration CI-5: unequal- N clusters free-fall identically while $F \propto N$); the collective-coordinate reduction is OPEN (FTD-0250).
8	EWSB / condensate	MEASURED	sharp first-order manifestation transition at amplitude $\in (0.6, 0.7)$; condensate mass gaps agree across the flux and charge channels to 3%.
9	Confinement signature	MEASURED	Wilson-loop area law with string tension $\sigma \approx 0.209$ at $\beta = x_-$; the coupling insertion is a SELECTION , not a derivation.
10	Classical electrodynamics suite	DERIVED	retarded radiation (FTD-0113), Bianchi identities (FTD-0114), boosted Coulomb + lattice Cherenkov pole (FTD-0115), extended sources, Larmor (FTD-0120) — the lattice ED corpus.
11	Ward identities	MEASURED	Gauss residual $\leq 10^{-8}$ (matched-stencil conjugate-gradient Poisson).
12	Gravity ratio	DERIVED (flagged step)	$(1\alpha_G(e, e) = \left(\frac{m_e}{m_P}\right)^2 \approx 1.745 \times 10^{-45}$ predicted vs 1.752×10^{-45} measured (0.38%), via Phase G + cluster mass + the clock hypothesis (the flagged AXIOM step); prediction floor STRONGLY MOTIVATED CONJECTURE (FTD-0131). Reproduced in §5.3.3.
13	Hydrogen spectrum	MEASURED	bound-state level structure $1/n^2$ to $< 0.001\%$ in the engine benchmark — but the benchmark places <i>classical Kepler</i> orbits at $r_n = n^2 a_0$ and checks the scaling, which is generic to any $1/r$ force; not a quantum eigenvalue derivation (see §5.5.1).
14	Bell bound	THEOREM	substrate $S \leq 2$ (FTD-0243 + Bell's theorem); engine $S \approx 1.95\text{--}2.00$; the $S \approx 3.6$ ternary-detection artifact is a known loophole, not a violation (Chapter 6, PL-2).
15	Detection statistics	CLOSED NEGATIVE	threshold-crossing event rates follow Rice's upcrossing law ($R^2 = 0.9923$), <i>not</i> Born $ J ^2$ ($R^2 = 0.7137$), in the pre-registered six-neighbour construction — CLOSED NEGATIVE for Born, NUMERICAL FACT for the Rice fit (FTD-0200; companion FTD-0199).

Three features of the catalog deserve emphasis before we reproduce its load-bearing rows. First, the tags are heterogeneous *by design*: rows 1, 2, 5 are theorem-grade; rows 3, 4, 8, 11, 13, 14 are measurements; row 6 carries three tags at once (linear DERIVED, bridge OPEN, identification STRONGLY MOTIVATED CONJECTURE); row 9 wears its inserted coupling as a SELECTION; row 15

is a CLOSED NEGATIVE for the Born rule. A single uniform tag over the catalog would be an overclaim. Second, several rows are zero-parameter consequences of lattice geometry (rows 1, 2, 5, 11), and these are the framework's strongest empirical content because there is nothing in them to tune. Third, the catalog deliberately *omits* α : the fine structure constant does not appear as an emergent IR measurement at all. It lives in the algebraic spine (Chapter 3) as a match, and §5.2 states exactly why that match is not promoted to a derivation.

5.2. The bridge: $x_+ = 1/\alpha$, and why it stays a conjecture

The master quadratic of Chapter 3,

$$P(x) = x^2 - 16 G^{*2} x + 16 G^{*3} = 0,$$

has the larger root

$$x_+ = 137.0361714582 \dots,$$

and the CODATA 2022 value of the inverse fine structure constant is

$$\alpha^{-1} = 137.035999177(21) \quad (\text{CODATA 2022}).$$

The agreement is

$$\frac{|x_+ - \alpha^{-1}|}{\alpha^{-1}} = 1.26 \times 10^{-6} = \mathbf{1.26} \text{ ppm},$$

roughly one part per million at tree level. This is the single most arresting number in the framework: a constant assembled from the lemniscatic bridge G^* (Chapter 3), with no electromagnetic input anywhere in its construction, landing on top of α^{-1} to sub-ppm precision.

THE CENTRAL CONJECTURE

The identification $x_+ = 1/\alpha$ is a STRONGLY MOTIVATED CONJECTURE (FTD-0013), a *match*, not a derivation. The polynomial itself, its discriminant, and its roots are THEOREM; the *physical reading* of the larger root as the inverse fine structure constant is conjectural. It is downgraded from an earlier THEOREM framing precisely because the structural-uniqueness evidence (FTD-0189: zero non- G^* dual-matchers across 2.65 million degree-2 polynomials over an 18-constant basket FTD did not design) strengthens the *evidence* for the conjecture without supplying a *derivation chain*.

The reason the identification cannot be promoted is the boundary mapped in Chapter 3. The route-invariance result (MC-T4.3, FTD-0242) found that across four independent FTD-native routes, *none forces* the operator assembly

$$(\text{Tr } W, \det W) = (16 G^{*2}, 16 G^{*3})$$

that the master quadratic encodes. The *menu* of that assembly is forced — the even trace $16 G^{*2}$ and the odd source G^* are genuine clean sources from the spine — but the *assembly* of those ingredients into the operator W whose characteristic polynomial is the master quadratic is not

compelled by any substrate route. In Chapter 3’s language: α is *dynamical, not structural*. That boundary is a FOUNDATIONAL OBSTRUCTION, NO-GO!, not a theorem and not a derivation of α ; and it is exactly what keeps $x_+ = 1/\alpha$ at STRONGLY MOTIVATED CONJECTURE!. The dashed arrow in the chapter mini-map (Figure 5.1), from the α -readout wall down to the $x_+ = 1/\alpha$ box, is the visual form of this dependency: remove the wall and the box could be solid; the wall is there, so it is dashed.

A second, independent reason reinforces the same verdict. The geometric-Coulomb result of §5.3.1 shows that when α *does* enter the engine’s emergent dynamics — through an explicit Gauss-source coupling $g_c = \sqrt{2\pi} \alpha$ — it does so as a *parametric insertion*, and the engine then reproduces QED Coulomb trivially. So the engine has exactly one place where α appears as an emergent number, and there it is inserted by hand; and exactly one place where α appears as a forced number, and there the forcing is conjectural. Neither place is a derivation of α from lattice dynamics. FTD has not demonstrated α as an emergent outcome of the substrate in any measurement to date, and this document does not claim otherwise.

5.3. Reproduced derivations

This section reproduces in full the three load-bearing chains of the physics layer: the geometric-Coulomb theorem (row 2), the cluster-mass coefficient $k = \frac{1}{N_{\text{base}}} = \frac{1}{4}$ at linear level (row 6), and the gravity ratio $\alpha_G(e, e)$ (row 12). Each uses the canonical macros; no constant is hand-keyed.

5.3.1. Phase-G geometric Coulomb is the lattice Poisson Green’s function

Claim 5.1 (Geometric Coulomb, THEOREM, FTD-0004). The engine’s emergent static pair potential has a closed-form prediction with *zero free parameters*: it is the periodic lattice Poisson Green’s function, carrying no fine-structure content whatsoever.

Setup. The engine’s Gauss law sources the flux field from the ternary state with no coupling constant in front: $\nabla \cdot J = \rho_s$, with $\rho_s = s \in \{-1, 0, +1\}$. The charge *is* the state value. On the L^3 cubic torus the seven-point Laplacian has the periodic Green’s function

$$G_L(r) = \frac{1}{L^3} \sum_{k \neq 0} \frac{e^{i k \cdot r}}{D(k)}, \quad D(k) = 2(3 - \cos k_x - \cos k_y - \cos k_z),$$

with $k_i = 2\pi n_i/L$ and the zero mode removed (charge neutrality on the torus).

Derivation. Place a +1 charge at n_1 and a −1 charge at n_2 . Each solves a discrete Poisson equation,

$$\nabla^2 \phi_1 = -\delta(n - n_1), \quad \nabla^2 \phi_2 = +\delta(n - n_2), \quad J_a = -\nabla \phi_a,$$

and by construction $\phi_1(n) = G_L(n - n_1)$. The pair flux is $J_{\text{pair}} = J_1 + J_2$, so the field energy contains the cross term $2 J_1 \cdot J_2$. Integrating by parts on the torus (no boundary),

$$\sum_n J_1 \cdot J_2 = \sum_n \nabla \phi_1 \cdot \nabla \phi_2 = - \sum_n \phi_1 \nabla^2 \phi_2 = \sum_n \phi_1 \delta(n - n_2) \cdot s_2 = -\phi_1(n_2) = -G_L(r),$$

with $r = |n_1 - n_2|$. The interaction energy is therefore $V(r) = -2 G_L(r)$, and the engine’s effective coupling readout is

$$\alpha_r(r, L) = 2 r G_L(r) = -V \cdot r$$

in the $\sum |J|^2$ energy convention.¹ For $r \ll L$ at large L the continuum limit of the lattice Green's function is $G_L(r) \approx 1/(4\pi r)$, so

$$\alpha_r(r) \approx \frac{1}{2\pi} \approx 0.1592,$$

a universal lattice constant depending only on r/L , with no α anywhere in it. A parameter-free FFT evaluation of $G_L(r)$ against the measured $\alpha_r(r, L)$ over $L \in \{32, \dots, 384\}$ reaches $R^2 = 1.0000$ with 0.07% median residual in the $L = 384$ Coulomb tail. $\square$

Reading. The lesson of this theorem is negative and important: the emergent static potential the engine produces is *geometric Coulomb* — pure lattice geometry, zero fine-structure content. The earlier “plateau at $3.6 \times \alpha_{\text{ref}}$ ” headline was a category error (a slice of $2r G_L(r)$ at fixed r/L , compared to an unrelated electroweak coupling). Where α enters the engine's dynamics it does so by an explicit inserted coupling $g_c = \sqrt{2\pi} \alpha$, under which the same theorem predicts $\alpha_r(r, L; g_c) = g_c^2 \cdot 2r G_L(r)$ and the engine reproduces QED Coulomb to lattice-spacing precision — a PARAMETRIC insertion, not an emergent derivation of α . This is the engine-side companion to §5.2's boundary.

5.3.2. The cluster-mass coefficient $k = \frac{1}{N_{\text{base}}} = \frac{1}{4}$ from O_h representation theory

Claim 5.2 (Cluster-mass linear coefficient, DERIVED at linear level, FTD-0110). The empirical cluster-size scaling $N(A) \approx \frac{1}{4} (A/K_{\text{genesis}})^2$ has its coefficient $k = \frac{1}{N_{\text{base}}} = \frac{1}{4}$ *derived* from the representation theory of the cubic point group O_h on the 27-voxel Moore block; the link to the full nonlinear engine is a STRONGLY MOTIVATED CONJECTURE, and a rigorous nonlinear bridge is OPEN.

The empirical fact. Engine campaigns measure $N(A) \approx k A^2$ with $k = 0.239 \pm 0.018$ across eleven amplitudes, five Standard-Model particles, and two injection geometries. The structural question is why $k \approx \frac{1}{4}$.

Step 1 — the 27-block decomposition (THEOREM). The 3^3 Moore block of 27 voxels carries the natural permutation representation ρ_{27} of O_h (order 48), with four orbits: the centre (1), the six SC face neighbours, the twelve FCC edge neighbours, and the eight BCC corner neighbours. It decomposes as

$$\rho_{27} \cong 4 A_{1g} \oplus 2 E_g \oplus 2 T_{2g} \oplus A_{2u} \oplus 3 T_{1u} \oplus T_{2u},$$

which checks dimensionally ($4 + 4 + 6 + 1 + 9 + 3 = 27$). The multiplicity of the trivial irrep follows from the character-table formula

$$\text{mult}(A_{1g}, \rho_{27}) = \frac{1}{|O_h|} \sum_{\text{classes}} (\text{size}) \chi_{\rho_{27}}(g) \chi_{A_{1g}}(g)^* = \frac{192}{48} = 4,$$

where $\chi_{\rho_{27}}(g)$ counts voxels fixed by g and $\chi_{A_{1g}} \equiv 1$. This is a theorem of group theory, independent of any FTD-specific machinery, and $\text{mult}(A_{1g}) = 4 = N_{\text{base}}$.

¹The current engine accumulates field energy with a $\frac{1}{2}$ prefactor (post-2026-04-27 refactor), which pairs $V(r) = -G_L(r)$ and $\alpha_r = r G_L(r)$; the form $\alpha_r(r, L) = 2r G_L(r)$ used here and in the IR catalog is the canonical $\sum |J|^2$ convention of FTD-0004, related by the factor of two (FTD-0286 v2). The physical content — that the readout is the lattice Green's function with no free parameter — is convention-independent.

Step 2 — the A_{1g} subspace and the projected Laplacian (**THEOREM**). The four-dimensional A_{1g} subspace is spanned by the orbit-averaged scalars

$$e_0 = \delta_{\text{centre}}, \quad e_1 = \frac{1}{\sqrt{6}} \mathbf{1}_{\text{SC}}, \quad e_2 = \frac{1}{\sqrt{12}} \mathbf{1}_{\text{FCC}}, \quad e_3 = \frac{1}{\sqrt{8}} \mathbf{1}_{\text{BCC}}.$$

The eighteen-point isotropic lattice Laplacian is O_h -equivariant, so by Schur’s lemma it preserves each irrep subspace; on $\text{span}\{e_0, e_1, e_2, e_3\}$ it acts as an explicit symmetric 4×4 matrix M whose four eigenvalues $\{-1.586, -3.862, -4.414, -4.805\}$ and eigenvectors are obtained by direct diagonalisation (trace $= -44/3$, confirmed against the full 27×27 operator to machine precision).

Step 3 — centre injection is A_{1g} -pure (geometric fact). The centre voxel is the unique fixed point of the O_h action: every cubic symmetry maps $(0, 0, 0)$ to itself. Hence $\delta_{\text{centre}} = e_0$ is O_h -invariant, so A_{1g} -pure, with zero overlap on any non-trivial irrep.

Step 4 — the mean energy fraction is $\frac{1}{4}$ (**DERIVED**). In the M -eigenbasis, δ_{centre} projects with squared coefficients

$$\left(\frac{3}{8}, \frac{1}{8}, \frac{3}{8}, \frac{1}{8}\right), \quad \sum = 1, \quad \text{mean} = \frac{1}{N_{\text{base}}} = \frac{1}{4}.$$

The mean is not a coincidence: it follows from two identities. The *sum identity* gives $\|\delta_{\text{centre}}\|^2 = 1$, and because the state is A_{1g} -pure that unit norm is carried entirely within the four A_{1g} eigenmodes. The *counting identity* gives exactly $N_{\text{base}} = 4$ such modes (Step 1). Therefore the mean per-mode energy fraction is $1/N_{\text{base}} = \frac{1}{4}$, *independently* of the non-uniform per-mode split $(\frac{3}{8}, \frac{1}{8}, \frac{3}{8}, \frac{1}{8})$ — which itself reflects the parity structure of the four eigenvectors and is washed out to equipartition under Langevin thermalisation.

Step 5 — from the mean to $N(A)$. The injected energy $E_{\text{inj}} = A^2 K_{\text{genesis}}^2$ distributes over the four A_{1g} modes; the cluster’s voxel count is the manifested energy divided by the per-voxel cost K_{genesis}^2 , harvesting the mean fraction:

$$N(A) = \frac{1}{N_{\text{base}}} A^2 = \frac{1}{4} A^2, \quad \text{i.e.} \quad N(A) \approx \frac{1}{4} (A/K_{\text{genesis}})^2.$$

The result is direction-invariant: an axial injection $J = (A, 0, 0) \delta_{\text{centre}}$ and a body-diagonal injection $J = (A/\sqrt{3})(1, 1, 1) \delta_{\text{centre}}$ give *identical* A_{1g} energy distributions $(\frac{3}{8}, \frac{1}{8}, \frac{3}{8}, \frac{1}{8}) A^2$ and the same mean $A^2/4$, because each Cartesian component evolves as an independent scalar field under the same scalar Laplacian. $\square$

Reading. Steps 1–4 are **THEOREM/DERIVED**: the character-table multiplicity, the projected eigenstructure, the A_{1g} -purity of the centre, and the mean-energy coefficient $\frac{1}{4}$ are all rigorous at the linear level. The *identification* of that coefficient with the full nonlinear engine’s cluster size — and hence with Standard-Model particle masses via $A = 2\sqrt{m_X/m_e}$ — is a **STRONGLY MOTIVATED CONJECTURE** (FTD-0110), supported empirically by 5 seeds $\times$ 11 amplitudes $\times$ 5 particles within the measured $k(A)$ drift envelope. The **OPEN** piece is to prove that genesis + evaporation + Langevin steady state reproduces the linear-mode equipartition; this is the cleanest remaining derivation gap of the physics layer.

5.3.3. The gravity ratio $\alpha_G(e, e)$

Claim 5.3 (Gravitational hierarchy, **DERIVED** (1 flagged step); floor **STRONGLY MOTIVATED CONJECTURE**, FTD-0131). The dimensionless gravitational coupling for one electron is predicted from the substrate as $\alpha_G(e, e) = \left(\frac{m_e}{m_P}\right)^2 \approx 1.745 \times 10^{-45}$, matching the measured 1.752×10^{-45} to 0.38%. The derivation chain is **DERIVED** modulo one flagged **AXIOM** step (the clock hypothesis); the prediction's epistemic floor is **STRONGLY MOTIVATED CONJECTURE**, inherited from the electron-mass formula it consumes.

*Step 1 — the form (**THEOREM**).* The discrete Poisson Green's function on the cubic lattice has the large- r tail $G_+(r) \rightarrow 1/(4\pi r)$ — the standard three-dimensional Newtonian form. This is the Phase-G result of §5.3.1 (FTD-0004), independent of any physics interpretation.

*Step 2 — the source (**DERIVED**).* An N -voxel cluster identifies with mass $M = N m_e$ under the calibration $K_B = m_e$ (cluster mass, row 6). Each manifested voxel acts as a gravitational source of the same strength, so the cluster's potential at large r is $\phi_g(r) = M K_B^{\text{grav}} / (4\pi m_e r)$, with $K_B^{\text{grav}} = K_B = m_e$ fixed by variation of the action (the manifestation term sources a Poisson equation $\nabla^2 \mathcal{L} = 4\pi G \rho$).

*Step 3 — the tick-rate response (**DERIVED** modulo the clock hypothesis).* The local tick rate responds linearly to the gravitational potential, $d\tau/dT_U = 1 + 2\phi_g/c^2$ (the metric component g_{00}); matching the leading Schwarzschild form $d\tau/dT_U = 1 - 2G_N M/(rc^2)$ identifies $G_N = -K_B^{\text{grav}}/(4\pi m_e)$. The square-root proper-time law is established by the action's Born–Infeld core; the single *flagged* **AXIOM** step is the clock hypothesis used in that identification — an interpretive step not yet substrate-derived.

Step 4 — the calibration-free prediction. The dimensionless ratio $\alpha_G(e, e) = G_N m_e^2/(\hbar c)$ is independent of any unit choice. By the Planck-mass relation $m_P^2 = \hbar c/G_N$,

$$\alpha_G(e, e) = \left(\frac{m_e}{m_P}\right)^2.$$

Substituting the electron-mass formula $m_e = m_P \sqrt{2\pi} \frac{16}{3} \alpha^{11}$ gives

$$\alpha_G(e, e) = \left(\frac{m_e}{m_P}\right)^2 \approx 1.745 \times 10^{-45}$$

against the measured 1.752×10^{-45} — a 0.38% match. Per error-propagation this is exactly twice the 0.19% precision of m_e/m_P itself; the squared form doubles the deviation. $\square$

Reading. The chain steps are **DERIVED** (modulo the one flagged **AXIOM**), but the prediction's load-bearing floor is **STRONGLY MOTIVATED CONJECTURE**; the 0.38% “match” is the squared precision of the electron-mass formula $m_e = m_P \sqrt{2\pi} \frac{16}{3} \alpha^{11}$, which is itself **STRONGLY MOTIVATED CONJECTURE** (its prefactor $\sqrt{2\pi} \cdot \frac{16}{3}$ and exponent 11 are motivated but not dynamically derived). The match is therefore *mechanical*, not independent new evidence; the substrate content is in the chain, not in the numerical agreement. Promoting $\alpha_G(e, e)$ to a clean **DERIVED** would require a first-principles derivation of that prefactor — which is **OPEN**. Independently, this derivation *falsifies* the older framework-integer reading $G_N = 1/(b_3 + N_c)^2 = \frac{1}{100}$ as an identification with physical G_N : it is off by twenty to forty-three orders of magnitude under every natural calibration, a numerical coincidence with no substrate justification.

5.4. The honest accounting

The physics layer’s integrity rests on a census, not a slogan. The parametric catalog (FTD-0254 lineage; canonical in the assessment ledger) enumerates ~162 quantities FTD reports for the Standard Model and sorts each by how it is obtained:

Category	Count	What it means
DERIVED / THEOREM	~23	formula <i>and</i> value produced from FTD axioms alone — genuine evidence
PARAMETRIC	~129	external formula (QED, ChPT, Fermi, Regge, seesaw) with FTD-supplied inputs — cross-check only
IMPOSED / SELECTION	~10	value fitted to experiment, or argued without a uniqueness proof — input, not output
Total	~162	

The reading is deliberately deflationary. A *derivation* takes the lattice axioms and produces both the functional form and the numerical value; a *parametric insertion* borrows the functional form from established physics and inserts FTD integers or couplings into it. Insertions are not errors — they are how any effective field theory cross-checks against the known Standard Model — but they are not independent evidence for FTD, and labelling them “derivations” would be the precise dishonesty the project’s anti-target discipline forbids. The ~23 in the left column are the load-bearing claims: the foundational constants (G^* , N_c , the master quadratic), the three lepton mass *ratios*, the geometric-Coulomb and mass-gap theorems, the classical-ED suite, the cross-sections, and the structural null-predictions. The ~129 in the middle column — the bulk of the meson and baryon masses, the decay widths, the precision-QED corrections — consume those few derivations as inputs.

This is the quantitative form of the intuition primer: the lattice reproduces a great deal of infrared physics, a small fraction of it is genuinely forced, and the framework states which fraction is which. The fine structure constant, the prize the whole construction seems to reach for, sits on the conjectural side of that ledger — matched to one part per million, never forced — and the accounting keeps it there.

5.5. What this layer does not settle

5.5.1. The hydrogen “spectrum” is a classical Kepler check

Row 13 of the catalog reports the hydrogen level structure $1/n^2$ to better than 0.001%, and the figure is real — but it is a **MEASURED** of a *classical* benchmark, not a quantum derivation. The engine places classical Kepler orbits at $r_n = n^2 a_0$ and verifies the $1/n^2$ scaling, which is generic to *any* $1/r$ central force; the Rydberg scale is calibrated through the inserted α and K_B . It is **not** a quantum eigenvalue derivation: there is no $\hbar$ in the substrate dynamics, no quantum kinetic operator, and no dynamical Pauli exclusion. Under FC-1 (Chapter 2) the framework formally *declines* the quantum-measurement layer, so atomic spectra are not, at present, substrate-derivable. The honest status of the atomic sector is that FTD derives essentially none of its quantum dynamics; the $1/n^2$ benchmark is a consistency check on a $1/r$ force, and should never be cited as more.

5.5.2. α is matched, not forced

The framework’s headline number, $x_+ = 1/\alpha$ to ~ 1.26 ppm (§5.2), remains a ~~‘STRONGLY MOTIVATED CONJECTURE’~~ (FTD-0013). The route-invariance boundary (MC-T4.3, FTD-0242) shows that no FTD-native route forces the operator assembly behind the master quadratic, so α is dynamical, not structural. This is the hinge of the whole document: the mathematics of Chapter 3 is forced up to a precisely located wall, and the physics of this chapter inherits that wall. Closing it would require a sixth-postulate-class input that forces the assembly, or a fresh engine-native measurement that derives α from dynamics — both ~~‘OPEN’~~.

5.5.3. Three derivation gaps and one imposition

Three of the catalog’s load-bearing rows carry explicit open kernels. The cluster-mass *nonlinear bridge* (row 6, FTD-0110) — proving that the nonlinear engine reproduces the linear-mode equipartition that delivers $k = \frac{1}{N_{\text{base}}} = \frac{1}{4}$ — is ~~‘OPEN’~~. The gravity ratio (row 12, FTD-0131) is ~~‘DERIVED’~~ only *modulo* the flagged clock-hypothesis ~~‘AXIOM’~~ step, and its prediction floor is ~~‘STRONGLY MOTIVATED CONJECTURE’~~; a first-principles derivation of the electron-mass prefactor $\sqrt{2\pi} \cdot \frac{16}{3}$ would lift it. Cluster transport inertia (row 7, FTD-0250) ships as an ~~‘IMPOSED’~~ engine model, with the collective-coordinate reduction — the dynamical twin of row 6’s bridge — left ~~‘OPEN’~~. None of these gaps is hidden; each is a mapped boundary of how far the discrete ontology, as run through the engine, currently reaches into infrared physics.

6. Science — Falsification

The move this layer makes: state in advance what kills you

A framework earns the name *science* not by accumulating agreements but by stating, before the experiment runs, what result would end it. The previous five layers built a structure: a finite ternary lattice (Ontology), a deterministic commutative logic that declines the quantum measurement import (Logic), a forced bridge constant and a forced quadratic with a precisely located wall (Mathematics), an observer layer that reads the lattice (Philosophy), and a computational effective field theory that *reproduces* infrared physics while only *matching* the one number everyone wants (Physics). This final layer turns the whole edifice around to face the world and asks: *where, exactly, does it break?*

The answer is the **deviation spine** — six structural deviations from the textbook QM/SR formalism, each carrying a discriminating protocol and an explicit kill condition. These are not parameter post-dictions; they are places where FTD’s native posture (a commutative substrate, declared by FC-1; a sector-scoped emergent metric, declared by FC-2) predicts that experiment will disagree with the textbook *structure*, not merely with a number. Crucially, this layer is also where the two framework declarations themselves are made killable: the constitution stakes in advance what would refute FC-1 and FC-2.

Read this chapter as a registry, not a result. Every row inherits its source’s tag; **inclusion in the spine never promotes a claim**. The deviation ledger adds discriminating protocols, scope caveats, and kill conditions — it adds no new physics. Two rows demand special honesty and are flagged as such below: the lab-Bell burden (PL-2), the framework’s highest-risk entry, and the proton-stability sub-claim of the structural nulls (PL-6), which a later audit (FTD-0301) demoted from a forced theorem to a selection.

6.1. The deviation spine PL-1 – PL-6

The spine is the registry FTD-0258 (a SYNTHESIS document; per-row tags live in the companion ledgers). Each row pairs an FTD result against the QM/SR expectation, names the discriminating protocol, states its scope, and gives the kill condition — the measurement that would falsify the FTD side. Table 6.1 reproduces the six rows at their source tags.

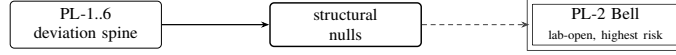


Figure 6.1.: This chapter in the chain. The science layer is where the construction becomes *accountable*: the deviation spine PL-1 – PL-6 collects the framework’s falsifiable structural departures from the QM/SR formalism, the structural nulls are **THEOREM**-grade absence claims, and the lab-Bell burden (PL-2) is the double-ruled wall — the framework’s single highest-risk entry.

Table 6.1.: The deviation spine: six structural deviations from the QM/SR formalism, each at its source tag. Inclusion never promotes; the row’s epistemic force is its source’s, not the registry’s.

Row	Deviation, protocol, kill condition	Tag / id
PL-1 Rice, not Born	Threshold-manifestation event frequency follows Rice’s Gaussian-upcrossing law ($R^2 = 0.9923$), beating the Born form $\propto J ^n$ ($R^2 = 0.7137$) by far above the pre-registered margin. <i>QM expects</i> the Born rule, detection probability $\propto \psi ^2$ exactly. <i>Protocol</i> : bin near-threshold detector rates by mean local intensity and fit Rice versus power-law on the same data; the signature is saturation of the rate envelope where Born predicts continued growth. <i>Scope</i> : measured in the 6-neighbour substrate at $L = 24$; the canonical 26-neighbour generalization is CONJECTURE (testable in v2). <i>Kill</i> : a P1 – P5 construction whose statistics fit Born better than Rice under the same protocol.	CLOSED NEGATIVE (Born) NUMERICAL FACT FTD-0200
PL-2 Bell / CHSH substrate bound	Any correlation experiment implementable inside the commutative substrate (no measurement-map M) is bounded $S \leq 2$; engine CHSH reads $S \approx 1.95$ – 2.00 . <i>QM expects</i> the Tsirelson value $S = 2\sqrt{2} \approx 2.828$ at optimal settings. <i>Protocol</i> : CHSH on raw substrate flux correlations, no complexification step — any substrate-native $S > 2$ outside the detection loophole kills FC-1. <i>Read the honesty block below</i> — this is the framework’s highest-risk row.	THEOREM MEASURED FTD-0243

Row	Deviation, protocol, kill condition	Tag / id
PL-3 Co-measurable quadratures	All quadratures of a mode are co-measurable: the symplectic pair has $\{q, p\} \neq 0$ (the winding is real) yet the observable commutator $[q, p] = 0$; measured leakage $\sim 10^{-16}$. <i>QM expects</i> conjugate incompatibility $[\hat{q}, \hat{p}] = i\hbar$. <i>Protocol</i> : simultaneous sharp readout of both quadratures without back-action; engine-native every tick, physically FC-1-conditional. <i>Kill</i> : a P1 – P5 derivation of quadrature incompatibility, or a demonstration that FC-1 epistemics cannot reproduce measured homodyne complementarity.	MEASURED FTD-0251
PL-4 Moving-clock rate	A moving wave-clock departs from exact γ by a calculable lattice UV correction vanishing as $\propto L^{-2}$ on the $\langle 100 \rangle$ axis at $v \lesssim 0.85$; at fixed L the clock over-dilates, bending <i>below</i> γ . <i>SR expects</i> exact γ at every scale and axis. <i>Protocol</i> : lead with the law, not with “ γ emerges” — fix the transverse mode and grow L ; the content is the L^{-2} convergence and the UV bend. <i>Scope</i> : scoped MEASURED ($\langle 100 \rangle$, moderate v); diagonals unconverged. <i>Kill</i> : the L^{-2} law breaking on $\langle 100 \rangle$, or exact γ where the lattice requires a correction — either kills FC-2’s metric half.	MEASURED (scoped) FTD-0252
PL-5 Isotropy	The substrate is anisotropic at UV and isotropizes in the IR with a measured power law: rotation-breaking residual exponent $p = 4.0008 \pm 0.0006$ ($\delta \propto k^4$), strongly irrelevant under Wilsonian counting. <i>SR expects</i> exact rotational invariance at every scale. <i>Protocol</i> : the closed-form anisotropy measurement; the cubic-anisotropy signature (an O_h fingerprint) distinguishes a lattice from rotationally-invariant Lorentz violation. <i>Kill</i> : the k^4 decay reversing or plateauing at larger L (isotropization stalling — kills FC-2’s emergent reading).	MEASURED FTD-0258

Row	Deviation, protocol, kill condition	Tag / id
PL-6 Structural nulls	The ontology <i>forbids</i>, all at source tag THEOREM: $\tau_{\text{proton}} =$ THEOREM ∞ (charge conservation exact via the Gauss constraint); $N_{\text{monopole}} = 0$ ($\nabla \cdot B \equiv \nabla \cdot \nabla \times J = 0$); $N_{\text{SUSY}} = 0$ (no fermionic grading on the ternary state space); extra dimensions = 0 ($\text{Aut}(E) ^2 = 2^D(D-1)!$ forces $D = 3$). <i>Competitors expect</i> proton decay (GUTs), monopoles, SUSY partners and extra dimensions (strings). <i>Protocol</i>: the world's existing null searches <i>are</i> the protocol. <i>Kill</i>: a single observed positive falsifies the corresponding theorem's physical reading outright. The $\tau_{\text{proton}} = \infty$ sub-claim is corrected by FTD-0301 — see §6.3.	FTD-0258

A standard caveat applies to every engine-native MEASURED row: the results are statements about the FTD substrate at the stated lattice size, stencil, and protocol; their *physical* readings are conditional on the calibration register ($a_{\text{phys}} \equiv \ell_P$, $M_{\text{REST}} = m_e$, t_{phys}), on the manifestation $\rightarrow$ detection mapping at its CONJECTURE-grade tag, and on FC-1/FC-2 as declared commitments. Dimensionless deviation *structure* (which law, which exponent) is calibration-invariant; absolute physical scales are not. Under $a_{\text{phys}} \equiv \ell_P$ the UV corrections of PL-4 and PL-5 are Planck-suppressed, so their laboratory reach is bounded by existing Lorentz-violation searches: the engine-native versions are testable now, the physical versions are long-horizon.

6.2. The lab-Bell burden: PL-2 read honestly

The single most important sentence in this chapter is that **loophole-free laboratory Bell experiments already measure $S > 2$** . PL-2 is therefore *not* a prediction that laboratory Bell tests will read 2.000. It is two things at once: a scoped substrate bound that is THEOREM-grade, and an accepted open burden that is the framework's most exposed flank. Softening either half would be dishonest; the boundary box states both at their canonical tags.

PL-2 LAB-BELL HONESTY BLOCK — THE HIGHEST-RISK ROW
The scoped bound (forced). Any correlation experiment fully implementable inside the commutative substrate — with no measurement-map M — satisfies realism, locality, and statistical independence, so Bell's theorem applies and bounds $S \leq 2$. This is THEOREM-grade via FTD-0243 plus Bell; the engine measures $S \approx 1.95$–2.00 across every tested configuration. (The apparent $S \approx 3.6$ at $\sim 49\%$ detection efficiency is the known detection loophole, an artifact, not a violation.) The open burden (accepted, not discharged). Real laboratories measure $S > 2$. FTD must therefore produce the observed $S > 2$ <i>at its observer layer</i>. The existing aggregate-correlation mechanism is SELECTION-grade, and the complexification it relies on is <i>exactly</i> an

instance of the measurement import M that FC-1 *declines*. So FTD's account of laboratory Bell violations is **OPEN**. This is the deviation spine's highest-risk entry and the most likely place the framework breaks.

Kill condition. Either a substrate-native (M-free) $S > 2$ beyond numerical error — which kills FC-1 directly — or a proof that *no* observer-layer mechanism consistent with FC-1 can yield the laboratory statistics, which turns the burden into a verdict against the framework's account of established experiments. (FTD-0258 registry; FC-1 declared by FTD-0255.)

6.3. The proton-stability correction (FTD-0301)

The structural-nulls row PL-6 stands at its source tag **THEOREM**: the four absence claims are theorems of the ontology as recorded in the constitution and the parametric catalog. But one of its four sub-claims — proton stability, $\tau_{\text{proton}} = \infty$ — has since been audited directly, and the audit **demotes that sub-claim**. The precedence rule applies without exception: **the LEDGER wins over the constitution**, so the corrected status governs.

PROTON STABILITY: $\tau_p = \infty$ IS A SELECTION, NOT A FORCED THEOREM

The forcedness audit FTD-0301 (**MEASURED**, UNFORCED-METASTABLE) attacked the claim that the discrete ontology *forces* $\tau_{\text{proton}} = \infty$. Its target was a four-proposition argument: (a) exact charge conservation, (b) the proton as a locked triad of three same-sign particles, (c) locked triads exempt from evaporation, (d) weak transmutation preserving triad structure. Two engine-free legs falsify the *forcing*:

- FTD has exactly one exact vector charge (the $U(1)$ sum $\sum s$) and *no* baryon or $B-L$ current. Baryon number is an emergent three-quark colour-singlet *cluster label*; conservation of $\sum s$ does not forbid $p \rightarrow e^+ + \pi^0$. Propositions (b)–(d) smuggle in a baryon law the framework never derives, and they contradict FTD's own baryogenesis sector.
- Source fact: the triad-binding lock fires only on *same-sign* triples, whereas the physical proton $uud = (+1, +1, -1)$ is mixed-sign. The lock therefore *cannot fire on a real proton*, so the evaporation-exemption of (c) is structurally inapplicable to its target object.

The run of record confirms the reading: a cold proton is metastable and never lock-protected; under heating with the weak channel on, 46/48 trials transmute $uud \rightarrow uuu$ (changing the charge), while the control transmutes 0/48 — falsifying proposition (d). **Net epistemic change:** the $\tau_{\text{proton}} = \infty$ tag is corrected **THEOREM** $\rightarrow$ **SELECTION** (emergent-metastable). The discrete ontology does not determine baryon stability, and its native dynamics *license* proton decay — a boundary in the spirit of mapping what discreteness does not reach. The other three structural nulls (monopoles, SUSY, extra dimensions) are **unaffected** and remain **THEOREM** at PL-6.

This correction does not weaken the falsifiability of PL-6 — it sharpens it. An observed proton

decay no longer would merely contradict a theorem; under the corrected reading the framework's own weak channel already supplies a decay mechanism, so the open empirical question is the *rate*, and the structural prediction $\tau_p = \infty$ is withdrawn as a forced statement.

6.4. Kill conditions for the framework commitments

The deviation spine makes the *predictions* killable; the constitution (§6.2) goes further and makes the two **AXIOM**-class *declarations* themselves killable, stated in advance. This is what distinguishes a framework commitment from an unfalsifiable stipulation: FC-1 and FC-2 are choices the theorem leaves open, but each comes with a published list of what would refute it.

FC-1 (the commutative substrate declines the measurement import M) — **AXIOM-class, FTD-0255 — is killed by any of:**

1. a substrate-native pair of observables with $[A, B] \neq 0$ derived from P1 – P5 (this would refute the commutativity-independence premise FTD-0243 and is the single most direct refutation of the constitution);
2. a P1 – P5 derivation of Born statistics (this would refute the binding-without-sharpness account and collapse the PL-1 deviation);
3. a substrate-native, M -free correlation experiment exceeding $S = 2$ beyond numerical error (the PL-2 kill condition).

FC-2 (native arrow; Lorentzian metric emergent-IR and sector-scoped; space $\perp$ time fundamental) — **AXIOM-class, FTD-0256 — is killed by any of:**

1. exact γ (or exact Lorentz invariance) at finite L off the IR limit — a measured *absence* of the predicted UV corrections where the lattice requires them (PL-4/PL-5 returning null at finite scale);
2. a P1 – P5 derivation of global reversibility (this would refute the FTD-0253 boundary);
3. failure of the measured IR approach — the L^{-2} law breaking on $\langle 100 \rangle$, or the k^4 anisotropy decay reversing: emergence claims that stop emerging.

FC-0 (the $\mathbb{Z}[i]$ reading) — **AXIOM-class, FTD-0254/FTD-0249 — is killed by** an inequivalent reading of the lattice symmetry that produces the same spine (uniqueness failure), or a spine theorem that fails under FC-0 (soundness failure).

A point of physical-scale honesty travels with the FC-2 conditions: under the calibration register the PL-4 and PL-5 UV effects are Planck-suppressed, so their laboratory accessibility is bounded by current Lorentz-violation searches. The engine-native versions are testable now; the physical versions are conditional and long-horizon. The ledger states both readings, and so does this chapter.

6.5. Methodology as institution

A list of kill conditions is only as good as the discipline that holds the project to them. FTD's epistemics are operational, not aspirational ([SYNTHESIS], constitution §6.3). Four institutional mechanisms make the falsification machinery binding rather than rhetorical:

Pre-registration. Every measurement campaign is hash-locked before it runs: a SHA-256 of the runner and the analysis, plus a git tag, recorded before any data exists (the manifest is the project's preregistration record). The verdict logic is frozen first; the run is the run of record. This is what prevents a near-miss from being retrofitted into a prediction — the deviation spine's L^{-2} blind extension and the time-dilation blind run are the recent exemplars.

The golden-tick gate. A bit-exact engine hash guards every physics-bearing change. An edit that perturbs the deterministic engine trace is caught immediately, so an instrument cannot silently drift between the claim and its check. Observation-only campaign code is proven golden-neutral (bit-identical hash with and without the counters) before its measurements count.

The LEDGER. A single source of truth with append-only tag history. Promotions happen only there, with evidence; no document — including the constitution and this one — may promote a claim. When a kill condition fires, the row is retagged in the LEDGER first, then annotated downstream. The precedence ordering LEDGER > constitution > prose makes the FTD-0301 proton correction (§6.3) binding on this chapter.

Anti-target discipline. No near-miss scans, no substitution identities, no labeling of a parametric insertion as a derivation. Honest negative results are first-class deliverables: the spine's PL-1 Born closure, the PL-2 open burden, and the FTD-0301 boundary are all recorded as such, not buried.

Together these turn the chapter's promises into commitments. The deviation spine says what disagreement would look like; the kill conditions say what would refute the declarations; and the institutional machinery guarantees that, when an experiment lands, the framework cannot quietly move the goalposts. This document was produced under those rules: it contains no new measurement and no promotion.

6.6. What this layer does not settle

The science layer maps the boundary of the framework's accountability; it does not pretend the boundary is everywhere closed.

- **The lab-Bell account is open** ([OPEN], FTD-0258 / FTD-0255). The substrate bound $S \leq 2$ is forced, but reproducing the measured laboratory $S > 2$ at the observer layer without smuggling back the declined measurement-map M is unsolved. This is the highest-risk entry, and it is genuinely unresolved, not merely awaiting data.
- **Several deviations are scoped, not yet universal.** PL-1 is measured in the 6-neighbour substrate; its canonical 26-neighbour reading is [CONJECTURE] (testable in v2). PL-4 is scoped [MEASURED] on $\langle 100 \rangle$ at moderate velocity, with the ultra-relativistic diagonals still unconverged

(OPEN). The structure is established where it is measured; its universality across the full configuration space is not.

- **The physical, as opposed to substrate-native, versions of PL-4/PL-5 are long-horizon.** Under the calibration register the UV corrections are Planck-suppressed; whether any laboratory can reach them is a question this layer poses but does not answer.
- **Kill conditions are stated, not yet exercised at the physical scale.** FC-1 and FC-2 carry published refutation lists, but most of those conditions can presently be checked only engine-natively. The framework has said what would end it; the experiments that could deliver the verdict, at the physical scale, are still to come.

A. Epistemic Accounting

This appendix records the bedrock against which the whole chain is measured. It reproduces the canonical truth-tier table and the ten entries that matter, and states the framework's honest claim/insertion accounting. The authoritative source is `TRACKER_ONTIC_TRUTH.md`; where this appendix and that tracker disagree on a tier, the tracker is correct.

A.1. The five truth tiers

Tier	Rating	Meaning
T1	★★★★★	Rock-solid: pure algebra or a three-line proof.
T2	★★★★	Conditional on a published classical theorem (rigour = the source's).
T3	★★★	Numerical fact verified over a stated domain (not a structural theorem).
T4	★★	Value-level coincidence; the structural reason is conjectured, not proved.
T5	★	Strongly motivated conjecture; includes empirical matches without a derivation chain.

A.2. The bedrock: the ten that matter

These are the entries one would defend to a skeptical mathematician. Each points at a verification artifact in the corpus.

ID	Result	Rating	Anchor
OT-1.1	Master quadratic + roots	★★★★★	FTD-0001
OT-1.2	G^* algebraic identity	★★★★★	FTD-0002
OT-1.3	Harmonic invariant tower	★★★★★	FTD-0111
OT-1.4	Phase-G geometric Coulomb	★★★★★	FTD-0004
OT-1.5	BCC complex-structure theorem	★★★★★	FTD-0122
OT-1.6	$\mathbb{Z}[i]^\times \rightarrow O_h^{ab}$ no-go	★★★★★	FTD-0122
OT-2.1	Watson identity	★★★★	Watson 1939 / Glasser–Zucker 1980
OT-2.2	Tower discriminant transcendence	★★★★	Schneider–Chudnovsky
OT-2.3	$\mathbb{Q}(G^*)$ is π -free	★★★★	Chudnovsky 1976
OT-3.4	Cluster coefficient $k = \frac{1}{4}$	★★★	FTD-0110

Two further ★★★★★ entries record alternative constructions of G^* (OT-1.7, the $\det_\zeta$ quarter-conjugacy bridge, FTD-0141; OT-1.8, the finite- N attractor, FTD-0142). The single ★ entry is the central conjecture itself: OT-5.1, $x_+ = 1/\alpha$ STRONGLY MOTIVATED CONJECTURE (FTD-0013, FTD-0242), which the framework explicitly does *not* claim as a theorem.

A.3. Claim/insertion accounting

The honest headline count of the framework, by canonical tag (CATALOG_PARAMETRIC_INSERTIONS.md):

Count	Category
~ 23	DERIVED / THEOREM — derived or theorem-grade.
~ 129	PARAMETRIC — standard physics formulas filled with FTD constants (mechanism borrowed).
~ 10	IMPOSED / SELECTION — calibration declarations and selection arguments.

The seven theorem-grade spine results plus the two $\mathbb{Z}[i]$ structural theorems and the cluster coefficient are the bedrock; everything else is downstream. The ratio is the project’s most important single fact about itself.

B. Provenance Registry

Every FTD-NNNN cited in this document, with a one-line description, as the reader's index back into the project ledger (LEDGER.md). Tags shown are the canonical tags; this registry promotes nothing.

Id	What it established	Tag
FTD-0001	Master quadratic $x^2 - 16G^{*2}x + 16G^{*3} = 0$ and its two real roots.	THEOREM
FTD-0002	The $G^* = \Gamma(1/4)/\Gamma(3/4)$ algebraic identity.	THEOREM
FTD-0003	CM-curve uniqueness among class-number-one fields.	THEOREM
FTD-0004	Phase-G geometric Coulomb = lattice Poisson Green's function.	THEOREM
FTD-0005	Phase-J partition-function ultralocality (at $L = 2$).	THEOREM
FTD-0006	Coefficient $16 = \text{Aut}(E) ^2$ (structural necessity conjectural).	THEOREM
FTD-0013	$x_+ = 1/\alpha$ to 1.26 ppm: the central identification.	STRONGLY MOTIVATED CONJECTURE
FTD-0044	Per-voxel mass gap from the manifestation threshold.	THEOREM
FTD-0059	$a_{\text{phys}} \equiv \ell_P$ calibration; non-derivability enforced.	SELECTION
FTD-0096	Mass-unit calibration interface (μ from ℓ_P).	SELECTION
FTD-0110	Cluster mass $N(A) \approx \frac{1}{4}(A/K)^2$, linear level; nonlinear bridge open.	DERIVED
FTD-0111	Harmonic invariant tower $1/y_+ + 1/y_- = 1$.	THEOREM
FTD-0112	Field-theoretic characterization of $\mathbb{Q}(G^*)$ as π -free.	THEOREM
FTD-0113	Retarded radiation on the lattice (Liénard–Wiechert).	DERIVED
FTD-0114	Lattice Bianchi identities and Hodge duality.	DERIVED
FTD-0115	Boosted Coulomb and the lattice Cherenkov pole.	DERIVED
FTD-0117	$G^* \approx 2.9587 \neq \varpi$ correction (the typo-bug fix).	NUMERICAL FACT
FTD-0120	Maxwell-exploit closure (Larmor, Cherenkov rate, source-half).	DERIVED

continued on next page

Id	What it established	Tag
FTD-0122	BCC complex-structure theorem; $\mathbb{Z}[i]$ unification (roles 2,4 no-go).	DERIVED
FTD-0131	Gravitational ratio $\alpha_G(e, e)$ derived; $G_N = 1/100$ falsified.	DERIVED
FTD-0141	G^* via the $\det_\zeta$ quarter-conjugacy bridge.	THEOREM
FTD-0142	G^* as a finite- N attractor with $O(1/N^2)$ rate.	THEOREM
FTD-0153	Math-first ontology: primitives to readouts to physics.	SYNTHESIS
FTD-0187	Born rule not derived; observer-layer account only.	SELECTION
FTD-0189	Look-elsewhere scan: master quadratic is the unique dual-matcher.	NUMERICAL FACT
FTD-0199	Detection-statistics construction (companion to the Rice test).	CLOSED NEGATIVE
FTD-0200	Rice upcrossing beats Born (PL-1); Born closed-negative.	CLOSED NEGATIVE
FTD-0225	A closed route to substrate non-commutativity.	CLOSED NEGATIVE
FTD-0226	A closed route to substrate non-commutativity.	CLOSED NEGATIVE
FTD-0227	Observer binding self-blind-spot (partial); sharpness open.	THEOREM
FTD-0228	A closed route to substrate non-commutativity.	CLOSED NEGATIVE
FTD-0242	MC-T4.3: α is dynamical, not structural (route-invariant).	FOUNDATIONAL OBSTRUCTION
FTD-0243	Commutativity-independence: $\{q, p\} \neq 0$ yet $[q, p] = 0$.	THEOREM
FTD-0244	K-BIND square-root wall, closed theorem-negative.	CLOSED NEGATIVE
FTD-0249	The FTD construction monograph.	SYNTHESIS
FTD-0250	Cluster transport inertia = $N \cdot M_{\text{REST}}$.	IMPOSED
FTD-0251	Native clock: co-measurable symplectic quadratures.	MEASURED
FTD-0252	Time dilation: IR departure from γ vanishes as L^{-2} .	MEASURED
FTD-0253	Causal cone forced; Lorentzian metric emergent-IR, sector-scoped.	THEOREM
FTD-0254	The framework constitution (the three-register charter).	SYNTHESIS
FTD-0255	FC-1: the measurement-map import M is declined.	AXIOM
FTD-0256	FC-2: native arrow; metric emergent; space $\perp$ time at base.	AXIOM

continued on next page

Provenance registry, continued

Id	What it established	Tag
FTD-0257	Two-orthogonal-fields ontology ($J \perp s$; nested (q, p)).	SYNTHESIS
FTD-0258	The structural deviation-prediction spine, PL-1 to PL-6.	SYNTHESIS
FTD-0286	Energy-convention factor-of-two in the geometric-Coulomb readout.	DERIVED
FTD-0301	Proton stability unforced-metastable; $\tau_p = \infty$ not a theorem.	SELECTION

C. Glossary

Flux field $J \in \mathbb{R}^3$

The continuous, dispositional layer: a vector field encoding potential energy density. Reversible in isolation; carries the wave dynamics.

State field $s \in \{-1, 0, +1\}$

The discrete, actual layer: the ternary manifestation at each lattice site. Finite and irreversible.

The two coupling channels

The only interactions between the orthogonal fields: *genesis/evaporation* (downward, $J \rightarrow s$: flux crossing the manifestation threshold projects a state) and *Gauss sourcing* (upward, $s \rightarrow J$: manifested states source flux through $\nabla \cdot J = \rho_s$).

Symplectic quadratures (q, p)

The amplitude and rate of the flux field; the substrate's only native dynamical angle. Poisson-nontrivial yet observable-commutative: $\{q, p\} \neq 0$ yet $[q, p] = 0$.

G^* (bridge constant)

$G^* = \frac{\Gamma(1/4)}{\Gamma(3/4)} = 2.95867511919\dots$ The lemniscatic constant that bridges the dispositional and actual layers. *Not* the lemniscate constant $\varpi \approx 2.62205755$ (FTD-0117).

Master quadratic

$P(x) = x^2 - 16 G^{*2} x + 16 G^{*3}$, with roots $x_+ \approx 137.0361714582\dots$ and $x_- \approx 3.024$.

FC-0

The framework commitment reading the lattice's order-4 symmetry as $\mathbb{Z}[i]$. An **AXIOM**-class declaration (FTD-0254).

FC-1

The commitment that the commutative observable algebra is complete; FTD declines the measurement-map import M . An **AXIOM**-class declaration (FTD-0255).

FC-2

The commitment that the arrow of time is native and the Lorentzian metric is an emergent IR property; space $\perp$ time at base. An **AXIOM**-class declaration (FTD-0256).

Framework integers $\{3, 4, 7, 13\}$

$N_c = 3$ (colours), $N_{\text{base}} = 4$, $b_3 = 7$, $N_{\text{eff}} = 13$. Outputs of the structure, used in the physics-layer formulas.

Epistemic tags

The honesty layer. **AXIOM** (postulate), **THEOREM** (proven), **DERIVED** (chain reproduced), **SELECTION** (consistency-argued), **SYNTHESIS** (synthesis), **PARAMETRIC** (formula borrowed), **STRONGLY MOTIVATED CONJECTURE** (strong evidence, no derivation), **CONJECTURE** (proposed), **IMPOSED** (calibration), **MEASURED** (measured in the engine), **OPEN** (open), **CLOSED NEGATIVE** (closed negative), **FOUNDATIONAL OBSTRUCTION** (foundational obstruction).

Sources & Further Reading

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